

Feature spreading, redundancy, and blocking behavior

Scott Nelson (UIUC) & Eric Baković (UCSD)

September 12, 2025

Abstract

We propose a typology of feature spreading and blocking patterns using Boolean Monadic Recursive Schemes (BMRS; Bhaskar et al. 2020, Chandlee & Jardine 2021). BMRS describe input-output maps as logical transductions using conditional IF . . . THEN . . . ELSE . . . syntax, uniting computational and substantive universals. Our typology emerges from varying redundancy conditions within BMRS feature spreading functions. When potential targets fail to receive or propagate harmonic features, redundancy conditions override harmony by blocking feature reception, halting propagation, or preventing both, depending on their type and scope. This offers a unified computational account of variation in feature spreading and blocking without traditional representational assumptions.

1 Introduction

We propose a typology of iterative feature spreading and blocking patterns using Boolean Monadic Recursive Schemes (BMRS; Bhaskar et al. 2020, Chandlee & Jardine 2021). BMRS describe phonological input-output maps as logical transductions using conditional IF . . . THEN . . . ELSE . . . syntax and, when they recurse in only one direction, are expressively equivalent to subsequential functions. As a phonological framework, BMRS have been argued to provide a formalism that can unite the *extensional* computational universals that have been discovered in work rooted in formal language theory (Johnson 1972, Kaplan & Kay 1994, Heinz 2018) as well as *intensional* substantive universals that have emerged from descriptive and theoretical phonological analysis (e.g. Greenberg 1966, Hyman 2008). While the BMRS syntax can encode any subsequential function, our aim with this work is to identify the computational structures (i.e., extension) and basic predicate types (i.e., intension) that underlie different types of phonological spreading patterns.¹ Our proposal thus builds upon recent work that relates different types of phonological generalizations to specific BMRS structures (Chandlee & Jardine 2021, Jardine & Oakden 2023, Nelson & Baković 2024a).

Typological characterizations of phonological generalizations can arise from different types of evidence and explanation.² The types of typological generalizations that we are interested in here are those that emerge directly via properties of the formal phonological grammar. Typological variation in phonological patterning has been explained by the grammar in different ways, but there is often a rearrangement of primitive elements. In rule-based theories, rule re-ordering can result in linguistic variation (e.g., Dialects A and B; Joos 1942, Chomsky & Halle 1968). In Optimality Theory and its related constraint-based theories, constraint re-ranking delimits the set of possible grammars/languages (Prince & Smolensky 1993/2004). The efficacy of rule re-ordering and constraint re-ranking to describe typological variation is of course also dependent on the content of the specific rules and constraints that are used, thus showing that this type of typological work is ultimately dependent on both structure and substance.

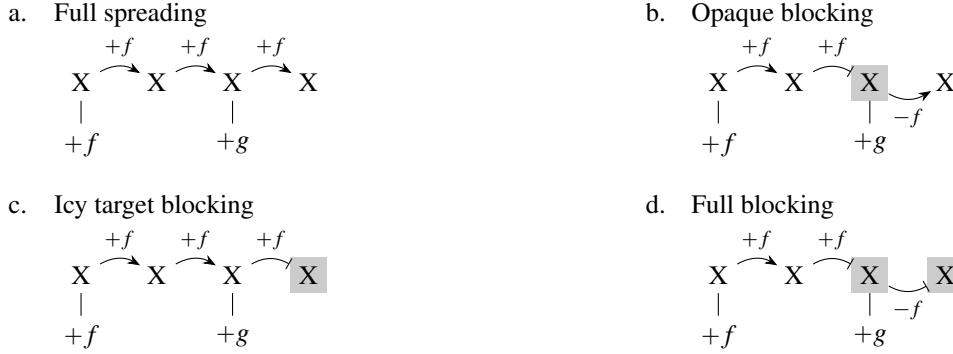
Typological work using BMRS is no different. Chandlee & Jardine (2021) provide a proof of concept for how BMRS can capture typological generalizations, demonstrating how Pater’s (1999) typology of *NC̥ repair strategies can be described by varying the use of predicates defining input sequences as licensing or blocking structures when computing nasality, voicing, and whether a given element is deleted in the output string that corresponds with a given input string. The content of these predicates is the substantive part of the repair strategy, and whether a given predicate licenses or blocks a given output property serves as the structural part necessary to describe a pattern within the typology. Our goal in this work is to describe different patterns of feature spreading and blocking rather than different repair strategies. This requires a slightly different approach to typological work using BMRS, which we will outline in §2 after providing an overview of the different types of patterns that we aim to capture with our typology.

¹ For slightly different perspectives on the role of computation in describing vowel harmony patterns, see Aksënova et al. (2024) and Burness et al. (2024).

² See the collection of papers in Hyman & Plank (2018) for a diverse, albeit non-exhaustive, set of approaches to phonological typology.

The schematic illustrations in (1) represent the four general types of feature spreading and blocking patterns that we focus on in this paper, highlighting the key differences among them. In all four cases, rightward spreading of $[+f]$ is shown to be initiated by the first element in the string, and the third element is flagged as a potential disruptor of this rightward spread. This is indicated with a $[+g]$ specification on this third element, standing in for whatever property of this element is responsible for this potential disruption. An arc with an arrowhead indicates that the indicated $[f]$ value succeeds in spreading from one element to another, the failure of spreading where it otherwise might be expected is indicated by a bar at the end of an arc, and the independence of an element from spreading is indicated by shading.

(1) Schematic typology of spreading and blocking types



The behavior of the third and fourth elements is what distinguishes each pattern from the others. In (1a), the third element both receives and further propagates the spread of $[+f]$ despite its in principle incompatible $[+g]$ specification; for obvious reasons, we refer to this type of pattern as FULL SPREADING. In (1b), the third element does not receive the spread of $[+f]$ and instead initiates its own spread of $[-f]$; this is the familiar type of OPAQUE BLOCKING pattern. The third element is shaded in this instance to indicate that its surface value of $[f]$ is independent of the spreading of $[+f]$ from its left. In (1c), the third element receives the spread of $[+f]$ but does not further propagate it. The third element here is dependent on spreading from its left, but the following fourth element is not, and is hence shaded. The third element in such a case has been dubbed an *icy target* (Jurgec 2011), so we refer to this type of pattern as ICY TARGET BLOCKING. In (1d), the third element neither receives the spread of $[+f]$ nor initiates its own spread of $[-f]$; we refer to this less familiar type of pattern as FULL BLOCKING. The surface values of both the third and fourth elements in this type of case are independent of the spreading from their left, again indicated with shading.

We derive the four patterns in (1), and variations on them, by varying the presence and scope of REDUNDANCY CONDITIONS within a BMRS harmony function. Redundancy conditions can either be INTERNAL, preventing the spread of $[+f]$ to the segment marked $[+g]$, or they can be EXTERNAL, preventing spreading to the segment *neighboring* the one marked $[+g]$. The basic idea is that when a potential target does not receive and/or further propagate the spreading feature, it's because the redundancy condition overrides spreading — and it does so in different ways, depending on its nature and scope within the larger harmony function.

Our assumptions are compatible with the *Richness of the Base* hypothesis of Optimality Theory in the sense that the only restriction on what is a possible input is that it must be a valid phonological representation; the absence on the surface of certain classes of representations emerges entirely from the input-output mapping. This idea expands upon the *representation as computation* approach to underspecification of Nelson & Baković (2024a,b), where it is shown that there are conditions on BMRS computation, including having the equivalent of a redundancy rule, that make the exact underlying specification of certain phonological features arbitrary. This arbitrariness of specification makes the representational underspecification analysis unnecessary, aligning with the spirit of Richness of the Base.

The remainder of the paper is structured as follows. In §2 we provide an overview of phonological computation with BMRS. We then begin §3 by describing how basic parameters of feature spreading and blocking (Archangeli & Pulleyblank 1994) correspond with different parts of BMRS functions. With these basic parameters in place, §3 continues with a demonstration of how full spreading (1a) and opaque blocking (1b) are accounted for, as well as subtle variants of these patterns that emerge from varying the scope of an internal redundancy condition. In §4 we complete our analysis of the typology by providing BMRS characterizations of icy target blocking (1c) and full blocking (1d), both of which involve external redundancy conditions. §5 discusses our typology and analyses as they relate to broader themes in phonological theories of spreading and blocking, and §6 concludes the paper.

2 Phonological computation as recursive schemes

A BMRS program defines string-to-string mappings via a collection of functions which determine whether a position in an output string carries a given property based on input properties of the string. All of these functions must take as their input a singular domain element (hence *monadic*) and have an output type of `Boolean` $\in \{\top, \perp\}$. Functions are built up using `IF . . . THEN . . . ELSE . . .` syntax, defined inductively as follows.

(2) Inductive definition of BMRS functions (Chandlee & Jardine 2021: 492)

- a. $\top$ (= true) and $\perp$ (= false) are expressions;
- b. any predicate $P(t)$ is an expression;
- c. if E_1 , E_2 , and E_3 are expressions, then so is ‘IF E_1 THEN E_2 ELSE E_3 ’;
- d. nothing else is an expression.

For example, the generic BMRS function in (3) determines whether a given position x in a string should have the property P in the output. This evaluates to $\top$ if x has the property Q in the input ($Q(x) = \top$) or, if it doesn’t have the property Q in the input ($Q(x) = \perp$), that it already had the property P in the input ($P(x) = \top$). The ϕ_P on the left side of the equation refers to an output property while the P and Q on the right side of the equation refer to input properties.

(3) $\phi_P(x) := \text{IF } Q(x) \text{ THEN } \top \text{ ELSE } P(x)$

Rather than generic P and Q , we can use phonological feature predicates $\mathcal{F} = \{\text{voice}, \text{nasal}, \text{sonorant}, \dots\}$ that are interpreted as binary, with values ‘+’ and ‘−’. If a given position x in a string has a given feature property f — i.e., if $f(x) = \top$ — then it is equivalent to saying that the element in that position is $[+f]$, and if it does not have that property (i.e., if $f(x) = \perp$) then it is $[-f]$. A somewhat less generic version of the function in (3) is given in (4), where P is replaced by the feature predicate for $[\text{voice}]$ and Q is replaced by the feature predicate for $[\text{sonorant}]$.

(4) $\phi_{\text{voice}}(x) := \text{IF } \text{sonorant}(x) \text{ THEN } \top \text{ ELSE } \text{voice}(x)$

The function in this case determines whether x should be $[+\text{voice}]$ in the output: this is true if x is $[+\text{sonorant}]$ in the input (i.e., sonorants are redundantly voiced), otherwise if it is $[+\text{voice}]$ in the input (i.e., obstruents contrast in terms of voicing). In terms that may be more familiar to phonologists, this function is in relevant respects equivalent to the input-output mapping encoded by the constraint ranking $\llbracket \text{SONORANTVOICE} \gg \text{IDENT(voice)} \rrbracket$ or by the context-independent redundancy rule $[+\text{sonorant}] \rightarrow [+\text{voice}]$.

2.1 Input and output context

The example BMRS function in (4) describes a context-independent input-output map, but most phonological maps are dependent on context. BMRS also includes `SUCCESSOR` $s(x)$ and `PREDECESSOR` $p(x)$ functions which provide a way to use the properties of elements in neighboring positions to determine a specific element’s output properties. For example, the function in (5) below nasalizes a vowel following a nasal. The process is non-iterative because the $[+\text{nasal}]$ property of the preceding element that determines nasalization is defined on the input.

(5) $\phi_{\text{nasal}}(x) := \text{IF } \text{syllabic}(x) \text{ THEN } \text{nasal}(p(x)) \text{ ELSE } \text{nasal}(x)$

This function determines nasality in the output by checking to see if an input element is a vowel (i.e., $\text{syllabic}(x) = \top$). If so, then that element will be nasal in the output only if the preceding element ($p(x)$) is a nasal in the input (i.e., $\text{nasal}(p(x)) = \top$). Any domain element that is not a vowel (i.e., $\text{syllabic}(x) = \perp$) will remain faithful to its input nasality, which is what the $\text{nasal}(x)$ predicate accomplishes in the ELSE clause. In order to make this vowel nasalization function iterative, we introduce a recursive call. Rather than determining the output nasality of a vowel based on the nasality of its *input* predecessor, we can include a recursive call and instead determine it based on its *output* predecessor. The resulting function, shown in (6), iteratively nasalizes all vowels following a nasal.³

³ We adopt the same “(innocuous) semantic convention” as Jardine & Oakden (2023: 2) “that, for any boolean function B , $B(s(x))$ $[/ B(p(x))]$ returns $\perp$ when $s(x)$ $[/ p(x)]$ is undefined—that is, x has no successor $[/$ predecessor].” This enables recursive output functions like (6) to terminate when there is no $\top$ element (an input nasal, in this case) to initiate the recursion; the recursive expression effectively ‘spreads’ $\perp$ from the domain edge.

$$(6) \quad \phi_{\text{nasal}}(x) := \text{IF } \text{syllabic}(x) \text{ THEN } \phi_{\text{nasal}}(p(x)) \text{ ELSE } \text{nasal}(x)$$

The evaluation tables in Figure 1 summarize how output nasality is computed for both non-iterative and iterative progressive nasalization. The first block of predicates indicates the input values that are necessary to compute the conditions in the function. Domain element 2 satisfies the condition of being $[+\text{nasal}]$ in the input. The output value $\phi_{\text{nasal}}(x)$ of this element is thus dependent on its input value $\text{nasal}(x)$, as highlighted by the leftmost red arrow in each evaluation table. The blue arrows indicate that the output value $\phi_{\text{nasal}}(x)$ of each of the domain elements 1, 3, and 4 is dependent in part on their satisfaction of the condition of being $[+\text{syllabic}]$ in the input. This output value is also dependent on the predecessor of each of these three domain elements. In the non-iterative case on the left, it is each predecessor’s input value ($\text{nasal}(p(x))$) that matters; in the iterative case on the right, it is its output value ($\phi_{\text{nasal}}(p(x))$) that matters. Domain element 1 lacks a predecessor in both cases; the semantic convention regarding domain edges (fn. 3) specifies this element as $[-\text{nasal}]$, as indicated by the dotted black loop arrows. In the non-iterative (left) case, the output $[\text{nasal}]$ value of each of domain elements 3 and 4 is determined by the *input* $[\text{nasal}]$ value of its predecessor, as indicated by the remaining red arrows, and in the iterative (right) case it is determined by the *output* $[\text{nasal}]$ value of its predecessor, as indicated by the dashed black arrows. Note that the recursivity of the iterative function in (6) is crucial for the determination of domain element 4’s output $[+\text{nasal}]$ value; its predecessor is $[-\text{nasal}]$ in the input, but $[+\text{nasal}]$ in the output due to spreading from *its* predecessor.

Input x	V 1	N 2	V 3	V 4
$\text{syllabic}(x)$	$\top$	$\perp$	$\top$	$\top$
$\text{nasal}(x)$	$\perp$	$\top$	$\perp$	$\perp$
$\phi_{\text{nasal}}(x)$	$\perp$	$\top$	$\top$	$\perp$
Output	V	N	$\tilde{V}$	V

Input x	V 1	N 2	V 3	V 4
$\text{syllabic}(x)$	$\top$	$\perp$	$\top$	$\top$
$\text{nasal}(x)$	$\perp$	$\top$	$\perp$	$\perp$
$\phi_{\text{nasal}}(x)$	$\perp$	$\top$	$\top$	$\top$
Output	V	N	$\tilde{V}$	$\tilde{V}$

Figure 1 Evaluation tables for non-iterative (left; see also (5)) and iterative (right; see also (6)) progressive nasalization. Arrows indicate the flow of computation and source of output values for each domain element.

2.2 Licensing and blocking structures

Whether a property evaluates to $\top$ or $\perp$ affects the overall evaluation of a BMRS function in a phonologically familiar way (Chandlee & Jardine 2021; see also Oakden 2021, Jardine & Oakden 2023). A **LICENSING STRUCTURE** has the form $\text{IF } P \text{ THEN } \top \text{ ELSE } \dots$, whereby satisfaction of an antecedent property P licenses the output property that is being computed to surface (i.e., ‘makes it true’). A **BLOCKING STRUCTURE** has the form $\text{IF } P \text{ THEN } \perp \text{ ELSE } \dots$, where non-satisfaction of P blocks the output property that is being computed from surfacing (‘makes it false’).

Licensing and blocking structures rely on composite properties with embedded $\text{IF} \dots \text{THEN} \dots \text{ELSE} \dots$ clauses. For example, we can take the non-iterative nasal spreading function in (5) and return an extensionally equivalent function with a licensing structure, (7a). The predicate $\text{NV}(x)$ in the antecedent of the licensing structure, defined in (7b), checks if domain element x has the property of being $[+\text{sonorant}]$ (a vowel) and the domain element to the left of x has the property of being $[+\text{nasal}]$ (a nasal) in the input. This structure licenses nasality at position x since satisfying it results in that position being specified as $[+\text{nasal}]$ in the output.

- (7) Non-iterative spreading (5) with a licensing structure
- $\phi_{\text{nasal}}(x) := \text{IF } \text{NV}(x) \text{ THEN } \top \text{ ELSE } \text{nasal}(x)$
 - $\text{NV}(x) := \text{IF } \text{syllabic}(x) \text{ THEN } \text{nasal}(p(x)) \text{ ELSE } \perp$

The functions in (7a) and (5) are extensionally equivalent, but they differ in terms of their intension. In (7a) we directly see that a sequence of a nasal followed by an oral vowel is a structure that licences nasality in the vowel: $\text{IF } \text{NV}(x) \text{ THEN } \top$ = ‘if x is $[+\text{syllabic}]$ and preceded by a $[+\text{nasal}]$ element, then x will also become $[+\text{nasal}]$ ’. In (5), on the other hand, this information can only be indirectly inferred: $\text{IF } \text{syllabic}(x) \text{ THEN } \text{nasal}(p(x))$ = ‘if x is $[+\text{syllabic}]$, then x will take on the $[\text{nasal}]$ value of its predecessor, whatever that may be’.

2.3 Faithfulness conditions

We distinguish *symmetric* spreading patterns, where both values of the harmonic feature spread from a specific source position (e.g. a root vowel or an edgemost vowel), from *asymmetric* spreading patterns, where one value of the harmonic feature is ‘dominant’ and spreads from wherever it may be in the input string. FAITHFULNESS CONDITIONS are another type of BMRS structure that can be used to account for symmetric spreading patterns. Rather than returning a default value ($\top/\perp$), faithfulness conditions ensure that a domain element retains its underlying value in the output.

Faithfulness conditions are of the form IF P THEN $f(x)$ ELSE ..., where $f(x)$ is the input predicate (= feature value) corresponding to the output function $\phi_f(x)$ being computed. These types of predicates have typically been placed in the final ELSE... clause of BMRS functions in order to describe what happens when a feature is neither licensed nor blocked. When placed in a THEN... clause for symmetric patterns, spreading becomes dependent on a value at a specific position (the source) rather than determined by a dominant value that may appear anywhere in the string.

Computing symmetric spreading patterns thus requires an extra step compared to asymmetric patterns. The source value of the spreading feature must be established in the case of symmetric patterns, but it is already defined in the case of asymmetric patterns. The consequence of this is that asymmetric patterns can be described using only licensing and blocking structures, while symmetric patterns may include a faithfulness condition to determine the source of spreading. As we’ll see, the placement of these faithfulness conditions relative to other components of a larger BMRS expression is meaningful, and effective in describing subtle variations in spreading and blocking patterns.

An obvious alternative here is to analyze symmetrical spreading as a special subcase of asymmetrical spreading: if the domain element in the source position is underlyingly specified with the dominant feature value, then its output value can be determined with a licensing or blocking structure, which can then be conditioned to spread; otherwise, the other value of the spreading feature is specified on all domain elements by default. We eschew this alternative if only to highlight the flexibility of our approach to accommodate either analysis of symmetrical spreading.

2.4 Summary

At its core, BMRS is simply a logical formalism that provides particular computational mechanisms for describing subsequential string-to-string functions. This echoes McCarthy’s (2008: 27) description of Optimality Theory as “a formal system in which notions like constraint priority are rigorously defined, but it does not say what the constraints are.” In the same way that Optimality Theory as a theory of phonology requires a (sub)theory of what is a possible markedness or faithfulness constraint, BMRS also requires a theory of possible licensing and blocking structures and faithfulness conditions separate from the computational system itself in order to be a full working theory of phonology. Our aim here is not to develop a complete theory of licensing and blocking structures for spreading patterns, but rather to elucidate the basic substantive elements underlying these types of structures. In the remainder of this paper we define a typology of iterative feature spreading patterns, deploying redundancy conditions *qua* licensing and blocking structures with varying scope within the functions determining the output value of the spreading feature.

3 Basic elements of feature spreading and blocking

The basic elements of the analysis of feature spreading and blocking are defined in (8), adapted from Archangeli & Pulleyblank (1994). We focus here on vowel harmony patterns, and assume throughout that the **domain** or fullest extent of spreading is the phonological word and that the only segments involved in the xpattern are vowels.⁴

(8) Basic elements of feature spreading and blocking

- a. **Content.** The phonological feature (value) that spreads within the domain.
- b. **Source.** The position in the domain that initiates spreading of the feature.
- c. **Orientation.** The direction in which the feature spreads, relative to the source.
- d. **Redundancy.** Conditions on the association of the feature with some positions.

⁴ Our functions therefore correspond to subparts of *tier-based strictly local* functions, which are subsequential (Burness et al. 2021). See Heinz et al. (2011) and De Santo & Graf (2019) for different approaches to computing tier strings from full string representations.

Each of these basic elements of spreading and blocking will be encoded in the vowel harmony functions examined in this section, as shown schematically in (9). Our focus is on the differences between functions that arise from the different embeddings (or complete absence) of IF...THEN...ELSE... expressions that encode **redundancy** (9d).

(9) Schematic feature spreading function subparts

- a. **Content determination.** $\phi_{\text{spf}}(x) := \dots$: The function determines the output value of the spreading feature [spf] for every domain element x .
- b. **Source faithfulness condition.** IF $\text{src}(x)$ THEN $\text{spf}(x)$ ELSE $\dots$: An x in the source position src is faithful to its input value of [spf].
- c. **Orientation setting (spreading condition).** $\dots$ ELSE $\phi_{\text{spf}}(\{\{s\}^p\}(x))$: An x not in src takes on the output [spf] value of its predecessor $p(x)$ or its successor $s(x)$.
- d. **Redundancy condition.** IF $\text{rdn}(x)$ THEN $\{\begin{smallmatrix} \top \\ \perp \end{smallmatrix}\}$ ELSE $\dots$: An x matching a condition rdn is redundantly specified as $[+\text{spf}] (= \top)$ or as $[-\text{spf}] (= \perp)$.

At this point it should be clear that redundancy conditions (9d) function either as licensing conditions, when the THEN... term is ' $\top$ ', or as blocking conditions, when it is ' $\perp$ '. We will see instances of both in the examples to be discussed in the remainder of this paper. For our purposes, however, the distinction between 'licensing redundancy' and 'blocking redundancy' is largely irrelevant: in both cases, one value of the spreading feature is prevented from spreading further — 'blocked', though in a different sense than that intended by the terms 'blocking structure'.

3.1 Full spreading

We begin our illustrations with a vowel harmony pattern with no redundancy conditions, Turkish backness harmony (Clements & Sezer 1982). The feature value assignments that we assume for the vowels of Turkish are given in (10).

(10) Turkish vowels and feature value assignments

	[-back]		[+back]	
	[-round]	[+round]	[-round]	[+round]
[+high]	i	y	ɯ	u
[-high]	e	ø	ɑ	o

In this case, the **content** (8a) is [back], the **source** (8b) is the (root-)initial vowel, the **orientation** (8c) is rightward, and there are no **redundancy** conditions (8d).⁵ Relevant examples are provided in (11). Roots are indicated with ' $\sqrt{}$ ' and boundaries between morphemes are indicated with '+'. The causative and past tense suffixes both have [+high] vowels and the conditional suffix has a [-high] vowel. The [+high] vowels alternate between [-back] [i] and [+back] [u] and the [-high] vowels alternate between [-back] [e] and [+back] [ɑ], in both cases ultimately depending on the [back] value of the preceding root vowel: [-back] in (11a,b) and [+back] in (11c,d).

(11) Turkish [back] harmony examples ($\sqrt{\text{root+CAUS+COND+PAST}}$)

- | | | | | | |
|----|--------------------------------|---------|----|-----------------------------------|---------|
| a. | $\sqrt{\text{gir+dir+sej+di}}$ | 'enter' | c. | $\sqrt{\text{kuur+duur+saj+duu}}$ | 'break' |
| b. | $\sqrt{\text{gel+dir+sej+di}}$ | 'come' | d. | $\sqrt{\text{atf+duur+saj+duu}}$ | 'open' |

This FULL SPREADING pattern corresponds to the simplest possible type of BMRS function in our typology.

(12) Turkish [back] harmony function — full spreading

$$\phi_{\text{back}}(x) := \text{IF initial}(x) \text{ THEN back}(x) \text{ ELSE } \phi_{\text{back}}(p(x))$$

⁵ Among [-high, -round] vowels in Turkish, [-back] [e] is redundantly [-low] and [+back] [ɑ] is redundantly [+low], leading these vowels to be harmonically 're-paired' with each other (Baković 2000). We set aside these types of redundancies because they are managed by the output functions for the non-spreading feature(s) involved ($\phi_{\text{low}}(x)$, in the case of Turkish).

The function in (12) states that the output [back] value of a given input vowel x ($= \phi_{\text{back}}(x)$) is the same as its input value ($= \text{back}(x)$) if x is initial ($= \text{initial}(x)$);⁶ otherwise, it is the same as the output value of x 's predecessor ($= \phi_{\text{back}}(p(x))$). The [back] **content** is expressed by the fact that the function determines the output specifications of [back] for all vowels. The spreading **source** — that the [back] value of all vowels is ultimately determined by (the faithful realization of) the underlying [back] value of an initial vowel — is expressed by the IF...THEN... clause of the function: IF the vowel is an initial vowel THEN its output value of [back] is the same as its input value. Finally, the rightward **orientation** of spreading is expressed by the ELSE... clause: the output value of any non-initial vowel is the same as the output value of the vowel that precedes it.

3.2 Opaque blocking

Adding a redundancy condition at the beginning of the full spreading type of function in (12) results in a function that describes an OPAQUE BLOCKING pattern. We illustrate this with ATR harmony in Standard Yoruba (Awobuluyi 1967). The feature value assignments that we assume for the vowels of Yoruba are given in (13); note the absence of [+high, -ATR] vowels from the inventory, indicated with ungrammaticality asterisks on the relevant symbols.

(13) Standard Yoruba vowels and feature value assignments

	[+ATR]		[-ATR]		
	[+high]	[-high]	[+high]	[-high]	
[-back]	i	e	*ɪ	ɛ	{ a } --- [+low]
[+back]	u	o	*ʊ	ɔ	----- [-low]

The **content** in this case is [ATR], the **source** is the final vowel, and the **orientation** is leftward. The **redundancy** condition (8d) is that all [+high] vowels are [+ATR], so [+high] vowels do not undergo [-ATR] spreading and block the further propagation of [-ATR]. Relevant examples are provided in (14). Examples (14a–f) illustrate full [ATR] spreading among [-high] vowels. Examples (14g–l) show that [+high] vowels do not undergo spreading of [-ATR], and examples (14j–l) show further that [+high] vowels initiate leftward spreading of their redundant [+ATR] value.

(14) Standard Yoruba [ATR] harmony examples

a. èrò	‘crowd’	g. ilɔ	‘house’
b. èfó	‘vegetable’	h. ifé	‘work’
c. ɛja	‘fish’	i. ilá	‘okra’
d. ògèdè	‘incantation’	j. òwúrò	‘morning’
e. ògèdè	‘banana/plantain’	k. òkígbe	‘magical drug’
f. ɛmèwà	‘chief’s messenger’	l. èḡìkà	‘shoulder’

This opaque blocking pattern corresponds to the following BMRS function.⁷

(15) Standard Yoruba [ATR] harmony function — opaque blocking

$$\phi_{\text{ATR}}(x) := \text{IF } \text{high}(x) \text{ THEN } \top \text{ ELSE} \\ \text{IF } \text{final}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE } \phi_{\text{ATR}}(s(x))$$

The harmony function in (15) states that the output [ATR] value of a given input vowel x ($= \phi_{\text{ATR}}(x)$) is [+ATR] ($= \top$) if x is [+high] ($= \text{high}(x)$); that otherwise it is the same as its input value ($= \text{ATR}(x)$) if x is a final vowel; and that otherwise it is the same as the output value of x 's successor ($= \phi_{\text{ATR}}(s(x))$). The redundant [+ATR] value of both final and non-final [+high] vowels is expressed by the top scope of the redundancy condition in the first IF...THEN... clause: IF any vowel is [+high], THEN its output value of [ATR] is [+ATR]. This redundancy condition is thus

⁶ Edge position predicates like $\text{initial}(x)$ and $\text{final}(x)$ are defined with reference to an adjacent domain boundary, ‘ $\bowtie$ ’ for domain-initial and ‘ $\bowtie$ ’ for domain-final. So, $\text{initial}(x) := \text{IF } \bowtie(p(x)) \text{ THEN } \top \text{ ELSE } \perp$, and $\text{final}(x) := \text{IF } \bowtie(s(x)) \text{ THEN } \top \text{ ELSE } \perp$.

⁷ The [+low, -ATR] vowel is similarly opaque in Standard Yoruba (e.g. [ɛrákpó] ‘type of plant’). This can be analytically accommodated in (15) by adding a redundancy condition with the same top scope as the one for high vowels: IF $\text{low}(x)$ THEN $\perp$ ELSE ...

technically a licensing structure, but as noted previously the distinction between licensing and blocking redundancy is irrelevant: either way, one of the harmonic feature values is prevented from spreading further. We will thus refrain from distinguishing whether a given redundancy condition is a licensing or a blocking structure in the remainder of this paper.

We have now seen an example of a harmony function without a redundancy condition — full spreading in the form of Turkish backness harmony, (12) — and an example of a harmony function with a redundancy condition with top scope — opaque blocking in the form of Standard Yoruba ATR harmony, (15). Two additional types of functions emerge simply from adjusting the scope of the redundancy condition, as we’ll now show.

3.3 Inventory reduction blocking

A different type of opaque blocking is exemplified by Turkish rounding harmony (Clements & Sezer 1982). The **content** is [round], and like backness harmony the **source** is the initial vowel and the **orientation** is rightward. There is also a **redundancy** condition: [−high] non-initial vowels are redundantly [−round], reducing the set of available vowel types from those available in the source, blocking the potential continued spread of [+round], and initiating [−round] spreading of their own. Relevant examples are provided in (16); (16e–h) are repeated from (11).

(16) Turkish rounding harmony examples ($\sqrt{\text{root+CAUS+COND+PAST}}$)

a.	$\sqrt{\text{gyl+dyr+sej+di}}$	‘laugh’	e.	$\sqrt{\text{gir+dir+sej+di}}$	‘enter’
b.	$\sqrt{\text{gør+dyr+sej+di}}$	‘see’	f.	$\sqrt{\text{gel+dir+sej+di}}$	‘come’
c.	$\sqrt{\text{dur+dur+sqj+duu}}$	‘stop’	g.	$\sqrt{\text{kur+dur+sqj+duu}}$	‘break’
d.	$\sqrt{\text{ol+dur+sqj+duu}}$	‘become’	h.	$\sqrt{\text{atf+dur+sqj+duu}}$	‘open’

The [+high] vowel of the causative suffix alternates between [−round] [i, u] and [+round] [y, ʊ], depending on the [round] value of the preceding initial vowel: [+round] in (16a–d) and [−round] in (16e–h). The [−high] vowel of the conditional suffix, however, only has the [−round] surface reflexes [e, a], blocking the continued spread of [+round]. Thus the [+high] vowel of the more peripheral past tense suffix still only has the [−round] reflexes [i, u] in (16), in agreement with the redundantly-specified [−round] vowel of the preceding conditional suffix.

We refer to this pattern as INVENTORY REDUCTION BLOCKING because the inventory of vowels that can surface in the source position is strictly reduced in non-source positions: among [−high] vowels, [+round] [o, ɔ] are only able to surface in the initial position. Otherwise, the impact of this inventory reduction on spreading is the same as opaque blocking: [+round] spreading is blocked by non-source [−high] vowels because they can only surface as [−round], and subsequent vowels surface as [−round] in agreement with the blocking vowel.

The difference between opaque blocking and inventory reduction blocking lies in the fact that the redundancy condition has top scope in the case of opaque blocking, affecting source and non-source vowels alike, while in the case of inventory reduction blocking the source faithfulness condition has top scope and thus the redundancy condition only affects non-source vowels. Compare the opaque blocking function for Yoruba [ATR] harmony with the inventory reduction BMRS function for Turkish rounding harmony in (17). Because the redundancy condition in the second line of this function requires reference to [−high] vowels, we define a predicate $[-\text{high}](x) := \text{IF } \text{high}(x) \text{ THEN } \perp \text{ ELSE } \top$ to directly reference such vowels in the harmony function rather than making it more indirect.⁸

(17) Turkish [round] harmony function — inventory reduction blocking

$$\begin{aligned} \phi_{\text{round}}(x) &:= \text{IF } \text{initial}(x) \text{ THEN } \text{round}(x) \text{ ELSE} \\ &\quad \text{IF } [-\text{high}](x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{round}}(s(x)) \end{aligned}$$

The harmony function in (17) states that the output [round] value of a given input vowel x ($= \phi_{\text{round}}(x)$) is the same as its input value ($= \text{round}(x)$) if x is an initial vowel ($= \text{initial}(x)$); otherwise it is [−round] ($= \perp$) if x is [−high] ($= [-\text{high}](x)$); and otherwise it is the same as the output value of x ’s predecessor ($= \phi_{\text{round}}(p(x))$).

⁸ If we were to make (extensionally equivalent) indirect reference to [−high] vowels, the second and third lines of the harmony function in (17) would be $\text{IF } \text{high}(x) \text{ THEN } \phi_{\text{round}}(p(x)) \text{ ELSE } \perp$. In other words, instead of directly stating ‘if x is [−high], then x is [−round]; otherwise, x agrees in [round] with its predecessor’, the function would state ‘if x is [+high], then x agrees in [round] with its predecessor; otherwise, x is [−round]’. The only difference is the relative placement of the redundancy condition result ($\perp$) and the spreading condition ($\phi_{\text{round}}(s(x))$), under THEN or under ELSE. This intensional difference is amplified when there are conditions involving multiple features; see fn. 13 below.

The [round] content is expressed by the fact that the function determines the output specifications of [round] for all vowels. The initial source of spreading is expressed by the first IF...THEN... clause of the function: IF the vowel is an initial vowel, THEN its output value of [round] is the same as its input value. The redundancy of [−high] vowels is expressed by the second IF...THEN... clause: IF the vowel is [−high], THEN its output value of [round] is [−round]. The fact that this redundancy condition only applies to non-initial vowels is expressed by the embedding of this second IF...THEN... clause after the determination of the source, ensuring that every initial vowel is faithful to its input [round] value. Finally, the rightward direction of spreading is expressed by the final ELSE... clause: the output value of any non-initial vowel is the same as the output value of the vowel that precedes it.

3.4 Inventory expansion spreading

It is also possible for a strictly reduced inventory of vowels to surface in the source position as compared to non-source positions. Because this is effectively the inverse of inventory reduction blocking, we refer to it as INVENTORY EXPANSION SPREADING.⁹ Inventory expansion spreading is exemplified by [ATR] harmony in the Ijeṣa dialect of Yoruba (Bamgboṣe 1967). Like Standard Yoruba, the **content** is [ATR], the **source** is the final vowel, and the **orientation** is leftward. The **redundancy** condition is also the same, except that only *final* (= source) [+high] vowels must be [+ATR]. The overall surface vowel inventory is thus the same as the one for Standard Yoruba in (13), except with the ungrammaticality asterisks on the [+high, −ATR] vowels removed. Relevant examples are provided in (18).

(18) Ijeṣa Yoruba (Bamgboṣe 1967)

a. ìgó	‘bottle’	d. ɪlá	‘okra’
b. ulɔ̌	‘house’	e. ʊsɛ	‘work’
c. ódiwèrè	‘he became a mad man’	f. ódɪbàbá	‘he became an old man’

Despite the absence of final [+high, −ATR] vowels, the non-final [+high] vowels alternate between [+ATR] [i, u] in (18a–c) and [−ATR] [ɪ, ʊ] in (18d–f), depending on the [ATR] value of the following final vowel. The example in (18f) show that non-final [+high] vowels further propagate the leftward spread of [−ATR]. This inventory expansion spreading pattern corresponds to the BMRS function in (19).

(19) Ijeṣa Yoruba [ATR] harmony function — inventory expansion spreading

$$\begin{aligned} \phi_{\text{ATR}}(x) := & \text{IF } \text{final}(x) \text{ THEN} \\ & \text{IF } \text{high}(x) \text{ THEN } \top \text{ ELSE } \text{ATR}(x) \\ & \text{ELSE } \phi_{\text{ATR}}(s(x)) \end{aligned}$$

This harmony function states that the output [ATR] value of a given input vowel x ($= \phi_{\text{ATR}}(x)$) is [+ATR] ($= \top$) if x is both final ($= \text{final}(x)$) and [+high] ($= \text{high}(x)$); that it is the same as its input value ($= \text{ATR}(x)$) if x is otherwise a final vowel; and that otherwise it is the same as the output value of x ’s successor ($= \phi_{\text{ATR}}(s(x))$). The [ATR] content of spreading is expressed by the fact that the function determines output specifications of [ATR]. The final source of spreading is expressed by the first IF...THEN... clause: IF the vowel is a root vowel, THEN the redundancy and faithfulness conditions apply. The redundancy of final [+high] vowels is expressed by the second IF...THEN... clause: IF the final vowel is [+high], THEN its output value is [+ATR]. The fact that this redundancy condition only applies to root vowels is expressed by the embedding of this second IF...THEN... clause before the otherwise faithful determination of the source. Finally, the leftward direction of spreading is expressed by the ELSE... clause: the output value of any non-final vowel is the same as the output value of the vowel that follows it.

3.5 Summary

The differences among opaque blocking, inventory reduction blocking, and inventory expansion spreading lie in the scope of the redundancy condition. It has top scope in the case of opaque blocking, determining the spreading feature value of redundant source and non-source vowels alike. In the case of inventory reduction blocking it has scope

⁹ Cases of this type have previously been referred to as *anti-structure preserving* (Baković 2000: 102ff; cf. Kiparsky 1981): a structure preserving process does not derive segment types that are not independently contrastive — or that do not also surface in the conditioning environment of the process, i.e. the source in the case of feature spreading — while an anti-structure preserving process does potentially derive such segments.

below the faithful determination of source vowels but above the harmonic determination of non-source vowels, thus determining the spreading feature value of redundant non-source vowels alone. In the case of inventory expansion spreading, the redundancy condition is embedded within the top THEN . . . clause, making it only relevant to the source and thus determining the spreading feature value of redundant source vowels alone.

Figure 2 graphs the flow of computation for and conversion relationships among these spreading patterns and the redundancy-free full spreading pattern. Full spreading can be converted to inventory reduction blocking by adding a

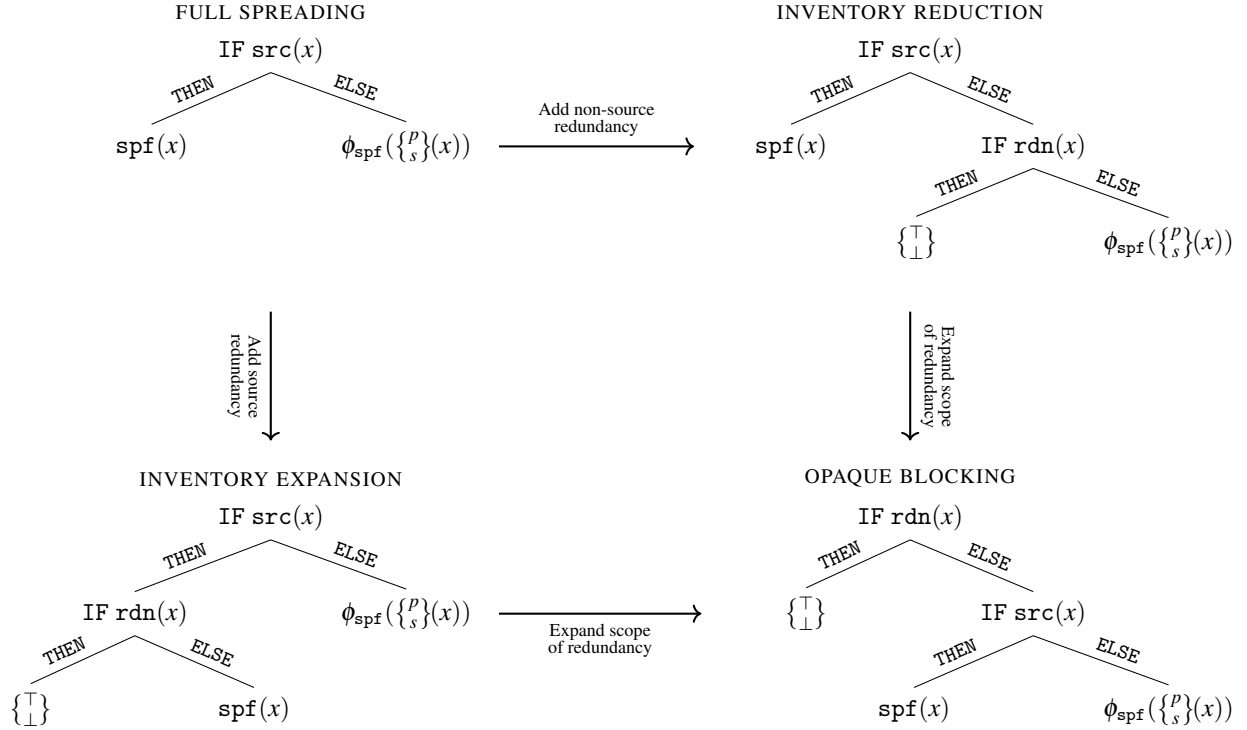


Figure 2 Flow of computation for — and conversion relationships among — full spreading, inventory reduction blocking, inventory expansion spreading, and opaque blocking patterns.

redundancy condition to the ELSE . . . clause, affecting only non-source vowels, or it can be converted to inventory expansion spreading by instead adding a redundancy condition to the THEN . . . clause, affecting only source vowels. Opaque blocking is the result of either adding a top-scope redundancy condition to full spreading or, equivalently, expanding the scope of the redundancy condition of either inventory reduction or inventory expansion spreading; having a single top-scope redundancy condition is extensionally equivalent to having both of the lower-scope redundancy conditions.

These four patterns instantiate two of the general spreading types illustrated in (1). Full spreading and inventory expansion spreading both correspond to the general pattern in (1a): in both cases, a non-source element receives and propagates the spreading feature regardless of any other feature specifications. Opaque blocking and inventory reduction blocking correspond to the general pattern in (1b): here, non-source elements that match the redundancy condition have their spreading feature value assigned by default. This allows them to be removed from the set of targets but still act as triggers for further propagation of the spreading feature. The distinction between patterns of type (1a) and (1b) therefore has to do with the presence or absence of a redundancy condition on non-source elements. The within-pattern distinction has to do with the presence or absence of a redundancy condition on source elements.

4 External redundancy

In the previous section we used redundancy conditions to describe two of the general types of spreading outlined in (1): full spreading (1a), of which inventory expansion spreading is a variant, and opaque blocking (1b), of which

inventory reduction blocking is a variant. Descriptions of the icy target blocking (1c) and full blocking (1d) patterns require redundancy conditions that reference not only the potential target of spreading, but also the adjacent segment on the side of the source. We refer to these as **EXTERNAL REDUNDANCY CONDITIONS**, schematized in (20b), as opposed to the **INTERNAL REDUNDANCY CONDITIONS** we’ve thus far been using ((20a), repeated from (9d)).

(20) Internal vs. external redundancy conditions

- a. **Internal redundancy.** IF $\text{rdn}(x)$ THEN $\left\{ \begin{smallmatrix} \top \\ \perp \end{smallmatrix} \right\}$ ELSE ... : An x matching a condition rdn is redundantly specified as $[+\text{spf}]$ ($= \top$) or as $[-\text{spf}]$ ($= \perp$).
- b. **External redundancy.** IF $\text{rdn}(\{s\}(x))$ THEN $\left\{ \begin{smallmatrix} \top \\ \perp \end{smallmatrix} \right\}$ ELSE ... : An x whose $\left\{ \begin{smallmatrix} \text{predecessor} \\ \text{successor} \end{smallmatrix} \right\}$ matches a condition rdn is redundantly specified as $[+\text{spf}]$ or $[-\text{spf}]$.

These two types of redundancy conditions have in common the fact that the output $[\text{spf}]$ value of a relevant segment x is not the result of $[\text{spf}]$ spreading from a neighboring vowel. In the case of internal redundancy (20a), x itself bears the property rdn that causes x to surface with a redundant value of $[\text{spf}]$. This is the sense in which the redundancy is *internal*: the critical piece of rdn information bearing on the output $[\text{spf}]$ value of x is found within x itself. In the case of external redundancy (20b), a neighboring vowel $p(x)$ or $s(x)$ bears the property rdn that causes x to be redundantly specified for a value of $[\text{spf}]$. This is the sense in which the redundancy is *external*: the critical piece of rdn information bearing on the output $[\text{spf}]$ value of x is found in a vowel neighboring x .

In the remainder of this section we discuss and analyze cases of icy target blocking and full blocking, using BMRS functions with external redundancy conditions, and thus complete our formal analysis of the full typology in (1).

4.1 Icy target blocking

An ICY TARGET BLOCKING pattern involves a set of vowels to which the harmonic feature can spread, but that do not themselves further propagate one of the values of the feature. Jurgec (2011: 91) named these kinds of vowels *icy targets* because they appear to ‘freeze’ spreading. We illustrate icy target blocking with dominant-recessive $[\text{ATR}]$ harmony in Iny (Ribeiro 2002, 2012).¹⁰ The feature values for the (oral, non-low) vowels of Iny are as given in (21).

(21) Iny vowels and feature value assignments

	$[\text{+ATR}]$		$[\text{−ATR}]$		
	$[\text{+high}]$	$[\text{−high}]$	$[\text{+high}]$	$[\text{−high}]$	
$[\text{−back}]$	i	e	ɪ	ɛ	$[\text{−round}]$
$[\text{+back}]$	ɨ	ɘ	ɨ	ɘ	$[\text{+round}]$
	u	o	ʊ	ɔ	

The **content** is $[\text{ATR}]$, the **source** is any underlyingly $[\text{+ATR}]$ vowel, and the **orientation** is leftward. Because $[\text{ATR}]$ harmony is dominant-recessive, any input vowel specified as $[\text{+ATR}]$ will be faithfully realized as $[\text{+ATR}]$ in the output with no chance of being overwritten. Icy target behavior is due to an **external redundancy** condition on potential triggers: an input $[\text{−ATR}]$ vowel is redundantly $[\text{−ATR}]$ (i.e., spreading is blocked) when it is followed by a $[\text{+high}, \text{−ATR}]$ vowel. The examples in (22) highlight these different aspects of the spreading pattern.¹¹

¹⁰ We follow McCollum (2022) and McCollum & Jardine (2022) in using the endonym *Iny* to refer to the language under discussion, as the name previously used in the literature is likely to be a racial epithet. See Ribeiro (2012: 7–8) for relevant discussion. Note that McCollum & Jardine (2022) also pursue a (more comprehensive) BMRS approach to $[\text{ATR}]$ harmony in Iny, arriving at a different analysis than we do here.

¹¹ The ‘=’ in these examples indicates an enclitic boundary. Note that the $[\text{+low}]$ vowel /a/ in example (22e) surfaces faithfully, presumably as $[\text{−ATR}]$ and thus disharmonically. This vowel and the $[\text{−high}]$ nasal vowels /ã, õ, ã/ are called ‘opaque’ by Ribeiro (2002, 2012), given that they all halt the leftward spread of $[\text{+ATR}]$ from a dominant vowel. Only the two $[\text{+low}]$ vowels /a, ă/ appear to be opaque in the usual sense, given that they are both (presumably) $[\text{−ATR}]$. The other two $[\text{+nasal}]$ vowels /ô, ô/ are both (also presumably) $[\text{+ATR}]$, which makes them something more like non-alternating icy targets. We unfortunately lack the space to properly address the issues raised by these facts, and assume that the analysis presented in the text is unaffected by them.

(22) Iny [ATR] harmony examples (Ribeiro 2002, 2012)

a.	$/\sqrt{kr\grave{o}br\dot{i}+d\ddot{i}}/$	$\longrightarrow$	$[kr\grave{o}br\dot{i}ni]$	‘a type of monkey’
b.	$/\sqrt{k\grave{o}d\ddot{u}+d\ddot{i}}/$	$\longrightarrow$	$[k\grave{o}d\ddot{u}ni]$	‘a type of turtle’
c.	$/\sqrt{h\grave{e}k\grave{o}d\ddot{i}+d\ddot{i}}/$	$\longrightarrow$	$[h\grave{e}k\grave{o}d\ddot{i}ni]$	‘oven’
d.	$/\sqrt{k\grave{o}l\grave{o}k\grave{o}+d\ddot{i}}/$	$\longrightarrow$	$[k\grave{o}l\grave{o}k\grave{o}ni]$	‘cedar’
e.	$/wa+\theta\epsilon+\sqrt{rik\grave{o}r\epsilon}/$	$\longrightarrow$	$[wa\theta\epsilon rit\grave{o}r\epsilon]$	‘my sibling’
f.	$/r+\grave{o}+\sqrt{t\dot{f}u\grave{h}\acute{o}=r\epsilon r\dot{i}}/$	$\longrightarrow$	$[rot\dot{f}u'h\acute{o}r\epsilon r\dot{i}]$	‘he is cursing’
g.	$/r+\grave{o}+\sqrt{t\dot{f}u\grave{h}\acute{o}=r+e}/$	$\longrightarrow$	$[rot\dot{f}u'h\acute{o}r\epsilon]$	‘he cursed’

The examples in (22a–d) show a single instance of leftward spreading of [+ATR] from the final suffix vowel to the root. In each instance, the root contains an icy target — an underlyingly [+high, –ATR] vowel, which undergoes spreading of [+ATR] but freezes its further propagation. Examples (22e–f) both show spreading from a root vowel onto a prefix vowel, and highlight the fact that any [+ATR] vowel initiates strictly leftward spreading, not just a final one. Example (22g) illustrates what happens when there are two input [+ATR] vowels, one final and the other medial: both spread leftward, and moreover, the medial [+high, +ATR] vowel does not behave as an icy target because it is underlyingly [+ATR], as can be appreciated by comparing this example with those in (22b–d).

The BMRS function in (23a) fully describes this complex pattern. First, the dominant-recessive nature of the harmony is captured by the fact that the predicates for the source and the spreading feature in the first line are identical ($= \text{ATR}(x)$). The second line thus determines the output of underlying [–ATR] vowels.¹² Because the external redundancy condition of this function references a [+high, –ATR] vowel, we define a predicate in (23b) to directly reference such vowels in the second line of (23a); recall the similar situation in the case of Turkish rounding harmony (fn. 8).¹³ If x ’s successor is [+high, –ATR], then x redundantly remains [–ATR] ($= \perp$). The output [ATR] value of all remaining vowels is determined by iterative leftward spreading ($= \phi_{\text{ATR}}(s(x))$).¹⁴

(23) Iny [ATR] harmony function — icy target blocking

- a. $\phi_{\text{ATR}}(x) := \text{IF } \text{ATR}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$
 $\quad \text{IF } \left[\begin{smallmatrix} +\text{high} \\ -\text{ATR} \end{smallmatrix} \right](s(x)) \text{ THEN } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x))$
- b. $\left[\begin{smallmatrix} +\text{high} \\ -\text{ATR} \end{smallmatrix} \right](x) := \text{IF } \text{high}(x) \text{ THEN IF } \text{ATR}(x) \text{ THEN } \perp \text{ ELSE } \top \text{ ELSE } \perp$

To underscore our analysis of icy target blocking in Iny, we include a BMRS evaluation table in Figure 3. This shows how the output [ATR] values of the vowels in $[h\grave{e}k\grave{o}d\ddot{i}ni]$ ‘oven’ (22c) are computed from the underlying [ATR] values of the vowels in $/\sqrt{h\grave{e}k\grave{o}d\ddot{i}+d\ddot{i}}/$. The first block of predicates indicates all of the input values that are necessary to compute the conditions in the actual function, which appear in the next block.¹⁵ Only domain element 4 satisfies the source faithfulness condition of being [+ATR] in the input. Its output value $\phi_{\text{ATR}}(x)$ is therefore dependent on its input value, as highlighted by the blue arrows. Only domain element 2 satisfies the external redundancy condition, meaning that its output [ATR] value is by default [–ATR], as highlighted by the red arrows. This is a faithfulness mapping in disguise because only input [–ATR] vowels end up being subject to the external redundancy condition, after the source faithfulness condition removes all [+ATR] vowels from consideration. The two remaining domain elements do not satisfy either condition and therefore are dependent on the following vowels’ *output* values for [ATR]. Domain element 3 precedes a [+ATR] vowel so this value is spread from domain element 4, causing a change in value from input to output. Domain element 1, on the other hand, precedes a [–ATR] vowel, which did not become [+ATR] due to the following icy target. Therefore, [–ATR] spreads from domain element 2 to domain element 1.

¹² Note that the final [–ATR] vowels in (22e–f) effectively surface as such not due to faithfulness but rather due to the semantic convention regarding domain edges (fn. 3). The *penultimate* [–ATR] vowels in these examples are handled either by spreading (22e) or by external redundancy (22f).

¹³ In this case, the very indirect alternative is for last two lines of (23a) to be $\text{IF } \text{high}(s(x)) \text{ THEN IF } \text{ATR}(s(x)) \text{ THEN } \top \text{ ELSE } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x))$.

¹⁴ Note that ‘THEN $\text{ATR}(x)$ ’ in the first line and ‘THEN $\perp$ ’ in the second line of (23a) could be substituted with ‘THEN $\top$ ’ and ‘THEN $\text{ATR}(x)$ ’, respectively; the result is in an extensionally equivalent function if either or both of these substitutions are made. The formulations of these function subparts in (23a) are consistent with parallel ones in previous functions, and with the schematic feature spreading function subparts in (9b).

¹⁵ Since the only underlying [+ATR] vowel of this example is also the final vowel, we skip the *final*(x) condition of the second line of the function.

Input	√ h	ε	k	ɔ	d	i + d	ĩ
x	1	2	3	4			
ATR(x)	⊥	⊥	⊥	⊥	⊥	⊥	⊥
$\begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(x)$	⊥	⊥	⊥	⊥	⊥	⊥	⊥
ATR(x)	⊥	⊥	⊥	⊥	⊥	⊥	⊥
$\begin{bmatrix} +\text{high} \\ -\text{ATR} \end{bmatrix}(s(x))$	⊥	⊥	⊥	⊥	⊥	⊥	⊥
$\phi_{\text{ATR}}(x)$	⊥	⊥	⊥	⊥	⊥	⊥	⊥
Output	h	ε	k	ɔ	d	i + d	ĩ

Figure 3 Evaluation table for $/\sqrt{h\epsilon k\omicron d\imath}+d\imath/ \rightarrow [h\epsilon k\omicron d\imath'ni]$ ‘oven’. Arrows indicate the flow of computation and source of output values for each domain element.

4.2 Full blocking

A FULL BLOCKING pattern is one in which a class of vowels neither receives nor propagates the spreading feature. We illustrate this type of pattern with rounding harmony in Halh (Khalkha) Mongolian (Svantesson et al. 2005). The feature values we assume for the short vowels in Halh are given in (24); long vowels are represented as doubled.¹⁶

- (24) Halh Mongolian vowels and feature value assignments

	[+ATR]		[-ATR]	
	[+high]	[-high]	[+high]	[-high]
$\begin{bmatrix} -\text{round} \\ +\text{round} \end{bmatrix}$	i	e	u	ɔ

The **content** is [round], the **source** is the initial vowel, and the **orientation** is rightward. There are two redundancy conditions in this case, one internal and the other external. The **internal redundancy** condition is that [+high] vowels (except for transparent /i/, which is not projected onto the harmony tier) are redundantly [-round]. The **external redundancy** condition is that a vowel following a [+high] vowel is redundantly [-round]. The [round] value of a [+high] vowel is thus neither determined by the preceding vowel nor determinant of following vowels — the hallmark of full blocking.

The examples in (25a–d) below illustrate how the [-high] vowel of the past tense suffix alternates to agree in [round] with the preceding root vowel. The causative suffix in examples (25e–h), on the other hand, is consistently realized with a [+round] vowel, even in examples (25f,h) where it is preceded by a [-round] vowel. The same [-high] vowel of the past tense suffix that was shown to alternate in examples (25a–d) is consistently realized with a [-round] vowel when preceded by the (equally consistently) [+high, +round] vowel of the causative suffix.

- (25) Halh Mongolian rounding harmony examples (root+PAST, root+CAUS+PAST)

a. $\sqrt{og}+\xi\omicron$	e. $\sqrt{og}+u\xi+\xi e$	‘give’
b. $\sqrt{xee\xi}+\xi e$	f. $\sqrt{xee\xi}+u\xi+\xi e$	‘decorate’
c. $\sqrt{or}+\xi\omicron$	g. $\sqrt{or}+u\xi+\xi a$	‘enter’
d. $\sqrt{jaw}+\xi a$	h. $\sqrt{jaw}+u\xi+\xi a$	‘go’

This full blocking pattern can be described with the BMRS function in (26).

¹⁶ Note that rounding and backness are mutually dependent; this is another case of re-pairing (recall fn. 5). The classification of this inventory differs from that in other works, e.g. Goldsmith (1985) and others cited by Svantesson et al. (2005: 222, Appendix E), who note that “[a]ll these assume palatal harmony and are therefore of limited value as analyses of Halh Mongolian.” Svantesson (1985, 2024) and Svantesson et al. (2005) analyze the alternations assumed in these works to be evidence of palatal harmony as *pharyngeal* harmony instead ([ATR] harmony, in our terms), which we do not address here. We also do not address the fact that /i/ is transparent to (both [ATR] and) [round] harmony in Halh. Since we already assume that vowel harmony functions operate over vowel tiers (recall fn. 4), one solution is simply to project all initial vowels, and all non-initial vowels except /i/, to the vowel harmony tier (*structure-sensitive tier projection*; De Santo & Graf 2019, Mayer & Major 2018). We leave a more comprehensive solution to the perennial problem of vowel transparency for future work.

- (26) Halh Mongolian [round] harmony function — full blocking
- $$\phi_{\text{round}}(x) := \text{IF } \text{high}(x) \text{ THEN } \top \text{ ELSE}$$
- $$\quad \text{IF } \text{initial}(x) \text{ THEN } \text{round}(x) \text{ ELSE}$$
- $$\quad \text{IF } \text{high}(p(x)) \text{ THEN } \perp \text{ ELSE } \phi_{\text{round}}(p(x))$$

The harmony function in (26) states that the output [round] value of a given input vowel x ($= \phi_{\text{round}}(x)$) is [+round] ($= \top$) if x is a [+high] vowel ($= \text{high}(x)$); otherwise, it is the same as its input value ($= \text{round}(x)$) if x is an initial vowel ($= \text{initial}(x)$); otherwise, it is [−round] ($= \perp$) if x 's predecessor is [+high] ($= \text{high}(p(x))$); and otherwise it is the same as the output value of x 's predecessor ($= \phi_{\text{round}}(p(x))$).

4.3 Expanding the typology

Recall from Figure 2 in §3.5 that the presence/absence and relative scope of internal redundancy conditions characterized the typology of spreading patterns discussed in §3. Our expansion of the typology to include the icy target blocking and full blocking patterns discussed in this section adds external redundancy conditions to the mix.

Each of the spreading patterns in the general typology in (1) is characterized by whether it includes certain atomic properties: a source faithfulness condition S which ensures that a vowel x in the source position is faithful to its underlying spreading feature value, an internal redundancy condition I which blocks a vowel x belonging to a particular class of vowels from receiving (a particular value of) the spreading feature, and an external redundancy condition E which blocks a neighboring vowel $\{p_s\}(x)$ belonging to a particular class of vowels from further propagating the spreading feature to x . In each case, spreading to x obtains only when these conditions are *not* met. Assuming that the source faithfulness condition S is required, there are four possible combinations of the presence/absence of the redundancy conditions I and E , corresponding directly with the four types of spreading patterns in (1).

- (27) Conditions required for the general typology of spreading patterns in (1)
- a. **Full spreading:** only S is present; spreading to x obtains only when $\neg S$ is satisfied (i.e., x is not a source).
 - b. **Opaque blocking:** both S and I are present; spreading to x obtains only when $\neg S \wedge \neg I$ is satisfied (i.e., x is not a source and also not redundantly specified for a particular value of spreading feature).
 - c. **Icy target blocking:** both S and E are present; spreading to x obtains only when $\neg S \wedge \neg E$ is satisfied (i.e., x is not a source and also not adjacent to a blocking vowel).
 - d. **Full blocking:** all of S , I , and E are present; spreading to x obtains only when $\neg S \wedge \neg I \wedge \neg E$ is satisfied (i.e., x is not a source, is not redundantly specified, and is not adjacent to a blocking vowel).

In §3 we showed how altering the position of the internal redundancy condition created its own subtypology within the space occupied by full spreading and opaque blocking. When it is embedded within the THEN . . . clause of the source faithfulness condition, the internal redundancy condition only restricts the set of vowels that can appear in the source position, which is already ineligible for spreading; inventory expansion spreading is thus a subcase of full spreading. When it is instead in the ELSE . . . clause, after the source faithfulness condition is computed, the internal redundancy condition instead only restricts the set of vowels that can appear in non-source positions; the result is inventory reduction blocking, a subcase of opaque blocking.

External redundancy conditions are different in this regard: given the external location of the property determining the redundant value, their interaction with the source faithfulness condition is not as direct. In Figure 4, we place the external redundancy condition below the source faithfulness condition in an effort to make the conversion relationships between connected patterns more visually apparent. Whether this positioning is necessary for the natural language examples discussed in this section varies. For Iny, the basic dominant-recessive nature of the harmony pattern means that the external redundancy condition must be lower because it relies on the higher source faithfulness condition removing all of the [+ATR] vowels from consideration. For Halh Mongolian, the position of the external redundancy condition relative to the source faithfulness condition makes no difference because the external determinant of the redundancy condition is (the [+high] value of) x 's predecessor, which is undefined for the initial vowel source.

Overall, Figure 4 shows the conversion relationships among the four main spreading patterns. Full spreading remains the most basic pattern where the only necessary check is whether a vowel is in the source position. Opaque blocking is obtained by adding a higher-scope internal redundancy condition, as discussed previously, because the redundantly specified vowels are unable to undergo spreading but can themselves further propagate the spreading

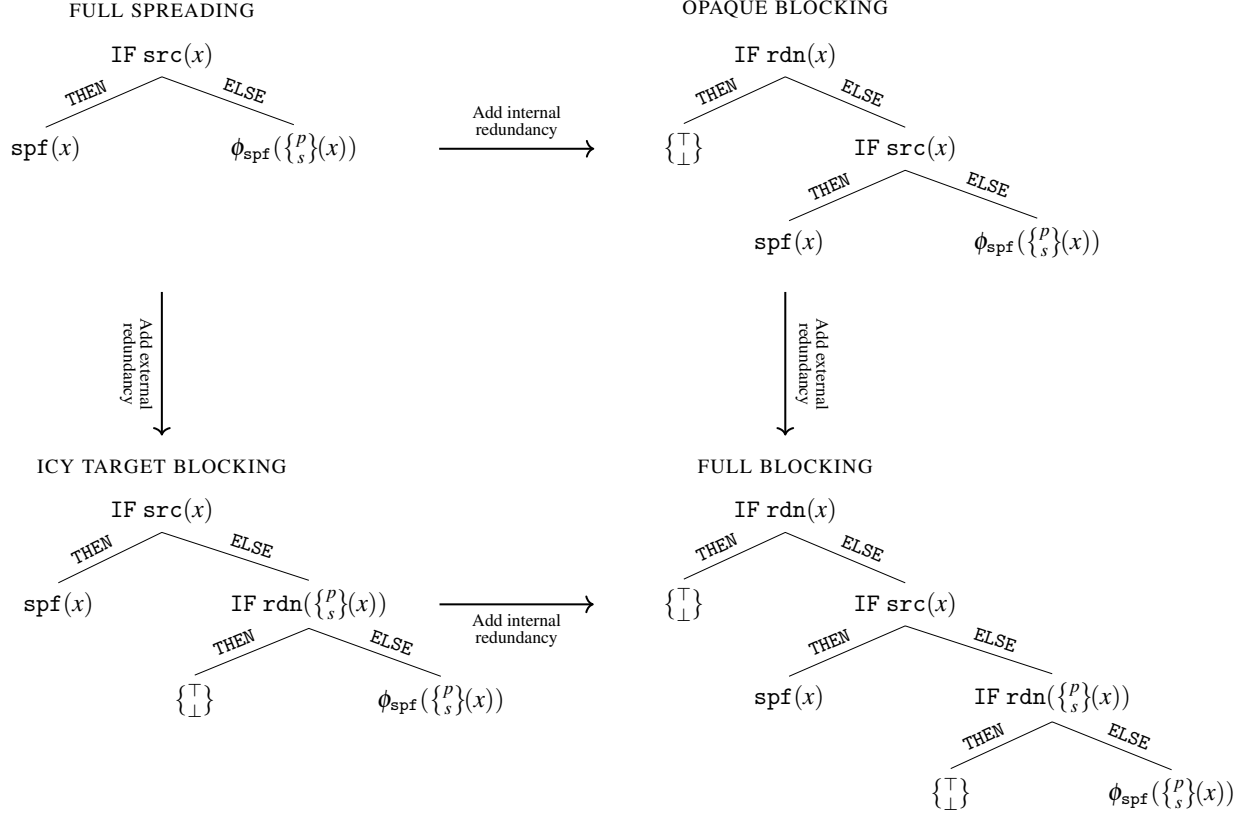


Figure 4 Flows of computation for — and conversion relationships among — full spreading, opaque blocking, icy target blocking, and full blocking patterns.

feature. If a lower-scope external redundancy condition is added to full spreading instead, the result is icy target blocking. This is because the external redundancy condition prevents a vowel from initiating spreading, but still allows it to be the recipient of spreading. Finally, adding the missing condition to either opaque blocking or icy target blocking results in full blocking. With three simple conditions, we have shown the relationship among these four types of general spreading and blocking patterns and what appear to be the core elements necessary to compute them.

5 Discussion

In this section we situate our approach within two broader themes in phonological theory. First, in §5.1, we discuss how our approach offers fundamentally computational solutions to problems that have traditionally been handled representationally. This discussion reveals a natural connection between our approach to feature spreading and blocking to recent BMRS work on underspecification. Then, in §5.2, we discuss feature spreading and blocking patterns that require access to information that is in principle unboundedly far from a given domain element not just in one consistent direction, but in either or both directions. Apart from the increase in expressivity of the computational function classes that these types of patterns require, they fit naturally into the typology we have thus far developed in this work.

5.1 Representation as computation

The approach to feature spreading and blocking patterns that we have developed in this paper re-characterizes what are traditionally thought of as *representational* properties of such patterns as *computational* properties instead. Consider, for example, the traditional representational account of feature spreading and blocking (see e.g. Clements 1981). In this account, what we have here been calling source faithfulness conditions and (internal) redundancy conditions are accounted for with the judicious deployment of feature prespecification and underspecification in the input.

- (28) Traditional representational account of source faithfulness and internal redundancy
- a. “To each regular root [i.e., the source position(s)] there corresponds one [spreading] feature, +F or –F. Affixes [i.e., non-source positions] are not specified for those features at any level.” (Clements 1981: 57–58)
 - b. “OPAQUE segments are just those segments that are already bound to some [value of the spreading feature], before the phonological rules (and the well-formedness condition) apply.” (Clements 1981: 61)

The well-formedness condition referenced in (28b), familiar from the autosegmental analysis of tone (Goldsmith 1976), is responsible for associating the spreading feature of the prespecified source and opaque vowels with the underspecified non-source vowels (Clements 1981: 45); opaque vowels block spreading from the source via the ‘association lines do not cross’ clause of this condition. The well-formedness condition is further supplemented with ‘priority clauses’ to prevent opaque vowels from spreading their prespecified feature values to vowels that are otherwise accessible to the source (Clements 1981: 62, 72), and language-particular adjustments must presumably also be made in order to describe the other types of blocking patterns discussed in §§3–4. While the graphical representation of spreading and blocking is striking and compelling, we contend that at least as much if not more of the descriptive and explanatory work is being accomplished by these essentially computational conditions on inputs and association.

Computation is of course ultimately dependent in some way on the representations that it operates over, so we also cannot entirely eschew representational concerns. What makes our approach to feature spreading and blocking computational rather than representational is that the locus of explanation falls squarely within the domain of the computation: whether the representation consists of *SPE*-like strings of attribute-value feature matrices or of autosegmental/feature-geometric graphs is completely irrelevant to the input-output map computed by our proposed BMRS harmony functions, so long as there is a ‘skeletal’ string of segments (or root nodes) and means by which to reference both the positions that they occupy and the featural properties that they have (or are associated with). The source faithfulness and redundancy conditions in our BMRS harmony functions handle the behavior of source and opaque vowels without the assumptions in (28) aimed at distinguishing these vowels from remaining non-source vowels in the input representation. Moreover, varying the position/scope of a redundancy condition — or its nature (internal vs. external) — produces the typology of attested spreading and blocking patterns depicted in §§3–4.

Our approach thus expands upon the one pursued by Nelson & Baković (2024a,b) to what they call UNDER-SPECIFICATION MAPS, for which they use BMRS to obviate the use of representational underspecification. These authors show that underspecification maps share a specific computational structure, making the choice of specification of putatively underspecified segments arbitrary. This arbitrariness implies that the inherently representational device of underspecification is not needed to analyze the relevant patterns, and that it can in fact lead to contradictory representational demands (Nelson & Baković 2024b, building upon observations by McCarthy & Taub 1992).

Nelson & Baković (2024a) discuss Russian voicing as an example case of an underspecification map. The relevant facts are that sonorants are always [+voice], obstruents are [–voice] domain-finally, obstruents contrast in terms of [voice] when followed by sonorants, and [voice] spreads regressively in strings of obstruents.¹⁷ A BMRS analysis of this underspecification map is provided in (29). The first line is the redundancy condition establishing that sonorants (= $\text{sonorant}(x)$) are redundantly [+voice] (= $\top$). The remaining lines thus determine the output voicing of obstruents. The second line states that any obstruent followed by a sonorant (= $\text{sonorant}(s(x))$) is faithful to its underlying [voice] value (= $\text{voice}(x)$). The last line begins with another redundancy condition establishing that word-final obstruents (= $\text{final}(x)$) are [–voice] (= $\perp$); otherwise, [voice] spreads regressively (= $\phi_{\text{voice}}(s(x))$).

- (29) Russian [voice] function
- $$\begin{aligned} \phi_{\text{voice}}(x) &:= \text{IF } \text{sonorant}(x) \text{ THEN } \top \text{ ELSE} \\ &\quad \text{IF } \text{sonorant}(s(x)) \text{ THEN } \text{voice}(x) \text{ ELSE} \\ &\quad \text{IF } \text{final}(x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{voice}}(s(x)) \end{aligned}$$

Nelson & Baković (2024a: 354) classify this as an underspecification map because it defines the input-output map for the putatively underspecified feature [voice], it contains a nested conditional BMRS term the upper condition of which (IF $\text{sonorant}(x)$ THEN $\top$) partitions the set of targets and the lower condition of which (IF $\text{sonorant}(s(x))$ THEN $\text{voice}(x)$) partitions the set of triggers, in both cases removing the class of segments identified by the antecedent of the corresponding redundancy rule in a traditional analysis ([+sonorant] $\rightarrow$ [+voice]). This analysis thus does

¹⁷ We set aside here the predictable voicing of /ts, tʃ, x/ (Halle 1959) as well as the ambiguous sonorant/obstruent behavior of /v/ (Padgett 2002).

not rely on underspecification: sonorants can be freely specified for [voice] in the input (i.e., *Richness of the Base*), and the function in (29) will reliably assign them their predictable output value of [+voice].

This non-reliance on underspecification is significant because, under a traditional representational approach, underspecification maps and opaque blocking patterns employ underspecification in very different ways. This is because the main class of redundantly specified segments in each case exhibit different behaviors: opaque vowels in opaque blocking patterns are triggers but not targets of spreading, while e.g. sonorants in the Russian underspecification map are neither triggers nor targets of anything but the redundancy rule that specifies them as [+voice]. Opaque vowels must thus be underlyingly specified for their redundant spreading feature value — or the relevant redundancy rule must apply early, “before the phonological rules (and the well-formedness condition) apply” (28b) — while sonorants in Russian must *not* be so specified, lest they trigger spreading of [voice]. In order for those same sonorants not to be spreading targets, there must be a separate statement that they cannot ever be specified as [−voice] (or that they cannot be specified for *either* value of [voice] before the redundancy rule that ultimately specifies them as [+voice]).

By contrast, there is no fundamental disparity between opaque vowels and Russian sonorants in our approach: both are redundantly specified via a (top-scope) internal redundancy condition, and hence neither is a spreading target. The fact that opaque vowels but not Russian sonorants are spreading triggers comes down to the content of the (lower-scope) source faithfulness condition: in Russian it is dependent on a neighboring segment of the same class as the internal redundancy condition (i.e., a following sonorant), while in the case of opaque vowels it is not.

To further reinforce the connection between this approach to underspecification maps and our approach to feature spreading and blocking, note that the Russian [voice] function in (29) is in many relevant structural respects equivalent to the kind of function that describes a full blocking pattern. This can be appreciated by comparing (29) with the Halh rounding harmony function in (26). In both cases, the first line of the function is an internal redundancy condition: in Russian it redundantly specifies sonorants as [+voice], while in Halh it redundantly specifies high vowels as [+round]. The second line in both cases is a source faithfulness condition: in Russian it ensures that obstruents before sonorants are faithful to their underlying [voice] value, while in Halh it ensures that initial vowels (i.e., those whose predecessor is a domain-initial boundary; recall fn. 6) are faithful to their underlying [round] value. The third line in both cases begins with an external redundancy condition: Russian final obstruents (i.e., those whose successor is a domain-final boundary) are redundantly specified as [−voice], while Halh non-high vowels whose predecessor is a [+high] vowel are redundantly specified as [−round]. The function ends in both cases with the spreading condition: regressive [voice] spreading in Russian, progressive [round] spreading in Halh.

Another notable similarity between the Russian and Halh functions is that the relative position of the source faithfulness condition and the external redundancy condition can be reversed in both cases (recall the discussion surrounding Figure 4 in §4.3). At this level of formal analysis, then, the most significant difference between these two functions is the fact that in Russian, internal redundancy and source faithfulness are both conditioned on the same feature ([sonorant]) — this is a key part of what makes this an underspecification map — while in Halh it is internal and external redundancy that are conditioned on the same feature (in that case, [high]). We leave the potential implications of the formal similarities and differences between these two functions to future research.

5.2 Bidirectional patterns

The recursivity of the spreading condition in a BMRS harmony function allows the computation to be informed by arbitrarily distant information. We have thus far only discussed patterns that require the use of UNIDIRECTIONAL recursive computation: patterns where feature spreading is strictly leftward or strictly rightward. Unidirectional recursion situates the functions describing these patterns within the class of DETERMINISTIC (SUBSEQUENTIAL) functions, those for which the output value of the relevant property/feature may be determined, for every domain element, by information that is either arbitrarily far to the left or arbitrarily far to the right of the domain element, but not both.¹⁸

Patterns that require BIDIRECTIONAL recursive computation have received heightened attention in the recent literature. These patterns can be divided into two groups, distinguished by the class of functions required to express them. The more expressive of these two classes is the NON-DETERMINISTIC (RATIONAL) class.¹⁹ Rational functions are those in which the output of at least some domain elements may be determined by information that is arbitrarily far on both sides. Patterns requiring this level of expressivity were dubbed UNBOUNDED CIRCUMAMBIENT by Jardine

¹⁸ The determining information may also be local to the domain element, whether in the element itself (by internal redundancy) or in a neighboring element (by external redundancy). This kind of local determination has no impact on the expressivity of a deterministic function.

¹⁹ This class has somewhat misleadingly been referred to as REGULAR in the literature; the rational functions technically constitute a SUBREGULAR function class. See fn. 1 of both Meinhardt et al. (2024) and Yolyan (2025) for recent corrections of the record on this point.

(2016). Functions in the less expressive class were dubbed **WEAKLY DETERMINISTIC** by Heinz & Lai (2013).²⁰ For these functions, the output of some domain elements may be determined by information that is arbitrarily far to the left, and that of other domain elements may be determined by information that is arbitrarily far to the right, but for no domain element can the output be determined by information that is arbitrarily far on both sides. Patterns requiring this level of expressivity were dubbed **UNBOUNDED SEMIAMBIENT** by Meinhardt et al. (2024).

The distinction between deterministic functions on the one hand, and both weakly deterministic and non-deterministic functions on the other, relates directly to recursive computation within BMRS. If recursion is only used in one direction, then the pattern can be classified as deterministic. If it requires recursion in both directions, then it belongs to one of the more expressive classes. While there are ways to extend the technical BMRS framework to capture bidirectional patterns (Bhaskar et al. 2023, Yolyan 2025), and thus to extend the expressivity of the formalism, we aim to show here that our typological characterization of feature spreading and blocking patterns remains intact.

5.2.1 Unbounded semiambiience

An example of a bidirectional, full spreading harmony pattern of the unbounded semiambiient type is found in the root-controlled [ATR] harmony pattern of Degema (Keri 1997, Archangeli & Pulleyblank 2007). As the examples in (30) show, both suffix and prefix vowels alternate in [ATR] agreement with the root vowel.

(30) Degema [ATR] harmony examples

- a. $\bar{u} + \sqrt{f\acute{o} + \bar{a}}$ ‘the state of being white’
- b. $\bar{u} + \sqrt{b\acute{o}m + \bar{a}m}$ ‘beating’
- c. $u + \sqrt{b\acute{i} + \bar{o}}$ ‘the state of being black’
- d. $u + \sqrt{d\acute{e}r + \bar{o}m}$ ‘cooking’

Another example of a bidirectional, full spreading harmony pattern of the unbounded semiambiient type is found in the dominant-recessive [ATR] harmony pattern of Diola Fogy (Sapir 1965, Ringen 1975). As the examples in (31) show, both [+ATR] root vowels (like those in /jitum/, ‘lead away’) and [+ATR] suffix vowels (like that in the ventive suffix /ul/) cause vowels to their left and right to alternate in [ATR] agreement with them.

(31) Diola Fogy [ATR] harmony examples

- a. $ni + \sqrt{baj + \epsilon n + \bar{u}}$ ‘I caused you to have’
- b. $ni + \sqrt{jitum + \epsilon n + u}$ ‘I caused you to be led away’
- c. $ni + \sqrt{b\acute{o}j + ul + u}$ ‘I have for you’
- d. $ni + \sqrt{jitum + ul + u}$ ‘I brought you’

Both of these patterns effectively consist of two unidirectional full spreading subpatterns, which can be described with the two functions in (32), one recursing in each direction. The difference between the root-controlled pattern in Degema and the dominant-recessive pattern in Diola Fogy is the content of the antecedent of the source faithfulness condition, $\text{src}(x)$: for Degema $\text{src} = \text{root}$, while for Diola Fogy $\text{src} = \text{ATR}$.

(32) Rightward and leftward [ATR] harmony functions — full spreading

- a. $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE } \phi_{\text{ATR}}(p(x))$
- b. $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE } \phi_{\text{ATR}}(s(x))$

Note that the ‘order’ in which these functions ‘apply’ — that is, whether they are composed such that the determination of vowels to the right of the source (32a) or the determination of vowels to the left of the source (32b) is prioritized — has no impact on the description of the pattern; the two functions may in fact apply simultaneously, because they don’t interact (McCollum et al. 2018, Meinhardt et al. 2021, 2024, Yolyan 2025). This is the hallmark of an unbounded semiambiient pattern, which can thus be described with a single weakly deterministic function. Two

²⁰ See also Graf (2016), McCollum et al. (2018, 2020), Hao & Andersson (2019), Lamont et al. (2019), O’Hara & Smith (2019), Smith & O’Hara (2019), Koser & Jardine (2020), Meinhardt et al. (2021, 2024), and Yolyan (2025) for further discussion of this class of functions.

alternative versions for a single BMRS function are provided in (33), following Jardine & Oakden (2023) and Yolyan (2025): one that prioritizes rightward spreading (33a) and another that prioritizes leftward spreading (33b). The extensional equivalence of these two functions mirrors the irrelevance of the ordering of the functions in (32).

(33) Unbounded semiambient [ATR] harmony functions — full spreading

- a. $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$
 $\text{IF } \phi_{\text{ATR}}(p(x)) \text{ THEN } \top \text{ ELSE } \phi_{\text{ATR}}(s(x))$
- b. $\phi_{\text{ATR}}(x) := \text{IF } \text{src}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$
 $\text{IF } \phi_{\text{ATR}}(s(x)) \text{ THEN } \top \text{ ELSE } \phi_{\text{ATR}}(p(x))$

As before, full spreading can be converted to other pattern types in our typology by adding redundancy conditions. Consider for example the results of placing the internal redundancy condition $\text{IF } \text{low}(x) \text{ THEN } \perp \text{ ELSE } \dots$ in various positions within the function in (33a), as summarized in Figure 5.

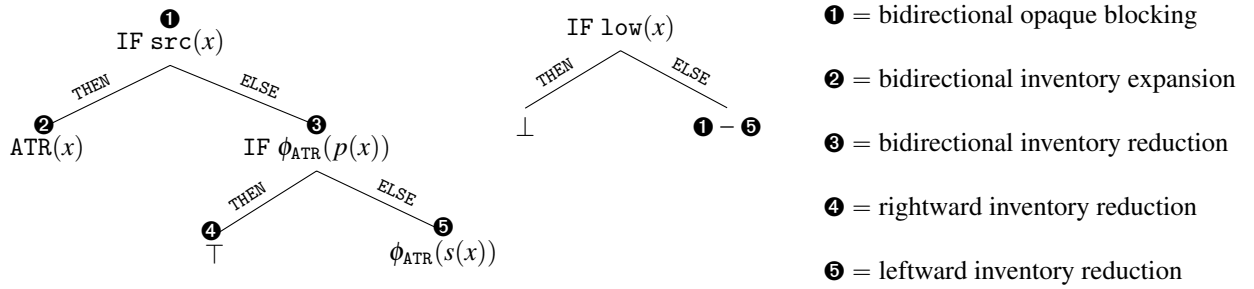


Figure 5 Results (right) of placing an internal redundancy condition (middle) in different positions within an unbounded semiambient harmony function (left). ❶ all low Vs are $[-\text{ATR}]$, ❷ src low Vs are $[-\text{ATR}]$, ❸ non- src low Vs are $[-\text{ATR}]$, ❹ right-of- src low Vs are $[-\text{ATR}]$, ❺ left-of- src low Vs are $[-\text{ATR}]$.

Giving the redundancy condition top scope (position ❶) describes a pattern where all low vowels are redundantly $[-\text{ATR}]$ and thus opaque blockers of $[\text{+ATR}]$ spreading in both directions. This matches the description of root-controlled [ATR] harmony in Akan found in Clements (1981) and Dolphyne (1988), though see now Casali (2012) and Kpogo (2024). If instead the redundancy condition only has scope over one of the two spreading conditions (positions ❹ or ❺), we describe a pattern with full spreading in one direction and blocking in the other. This matches the description of dominant-recessive [ATR] harmony in Maasai (Archangeli & Pulleyblank 1994, Baković 2000, Quinn-Wriedt 2013): low vowels block leftward but not rightward spreading of $[\text{+ATR}]$. It also matches the description of emphasis spread in Palestinian Arabic (Davis 1995, McCarthy 1997, Jardine & Oakden 2023), where high front segments block rightward but not leftward spread of emphasis/pharyngealization (identified as privative [RTR]).

Much more descriptive and analytical work remains to be done here, of course. For example, as made clear in Figure 5, the two unidirectional blocking patterns described by placing the redundancy condition in positions ❹ or ❺ are *inventory reduction blocking* patterns. Like $[-\text{high}]$ vowels in the Turkish rounding harmony pattern, the expectation is that vowels in the source position are not subject to the redundancy condition. A more complete description of the unidirectional blocking patterns in Maasai and Palestinian Arabic thus requires that something be said about why the relevant class of segments — low $[\text{+ATR}]$ vowels in Maasai; high front [RTR] segments in Palestinian Arabic — are not possible spreading sources. In the case of Maasai, we can appeal to the fact that $[\text{+low}, -\text{round}, -\text{ATR}]$ [ɑ] is ‘re-paired’ with $[-\text{low}, +\text{round}, +\text{ATR}]$ [o] under the influence of rightward spreading of $[\text{+ATR}]$. This fact would be captured by separate functions determining the output values of [low] and/or [round] with respect to [ATR], which would independently account for the lack of $[\text{+low}, +\text{ATR}]$ vowels as spreading sources as well. Things are much less straightforward for Palestinian Arabic, however: high front segments fully undergo *leftward* spreading of [RTR], so there would have to be a separate story told about why they are not also possible sources.

Patterns matching functions with the redundancy condition in positions ❷ and ❸ are not known to us at this time, nor are bidirectional patterns that can be augmented with external redundancy conditions to achieve icy target blocking or full blocking. Now armed with these analytical possibilities, we look forward to uncovering such patterns.

5.2.2 Unbounded circumambience

Tutrugbu has leftward [ATR] harmony, spreading from the root to prefixes (McCollum & Essegbey 2018, McCollum et al. 2018, 2020). There is also a blocking condition, crucially dependent on information at the other end of the word: a $[-\text{high}]$ vowel blocks the spread of $[+\text{ATR}]$ if the initial prefix vowel is $[+\text{high}]$. Examples are given in (34).²¹

(34) Tutrugbu [ATR] harmony examples

- | | | |
|----|---------------------|---------------------------------|
| a. | bu+tí+√fē | ‘1P+NEG+√grow’ |
| b. | e+tí+be+√fē | ‘3S+NEG+FUT+√grow’ |
| c. | bū+tí+√bá | ‘1P+NEG+√come’ |
| d. | a+tí+ba+√bá | ‘3S+NEG+FUT+√come’ |
| e. | i+dí+√wu | ‘1S+ITV+√climb’ |
| f. | bū+ba+di+√wu | ‘1P+FUT+ITV+√climb’ |
| g. | i+tí+ka+á+ba+ba+√wu | ‘1S+FUT+PFV+PRG-VNT+VNT+√climb’ |
| h. | e+tí+ke+é+be+be+√wu | ‘3S+NEG+PFV+PRG+VNT+VNT+√climb’ |

This pattern is unbounded circumambient because in order to determine the surface [ATR] value of a non-root vowel x it is necessary to know whether the root vowel, which can be unboundedly far to right of x , is $[+\text{ATR}]$; whether the initial prefix vowel, which can be unboundedly far to the left of x , is $[+\text{high}]$; and whether x itself is a $[-\text{high}]$ vowel or followed by a $[-\text{high}]$ vowel, which like the root vowel can be unboundedly far to the right of x .

Despite requiring information at a potentially unbounded distance in both directions, the Tutrugbu pattern is effectively a special instance of inventory reduction blocking: an internal redundancy condition specifying prefix $[-\text{high}]$ vowels as $[-\text{ATR}]$ is embedded below the source faithfulness condition, which allows $[-\text{high}, +\text{ATR}]$ vowels in the root. What distinguishes this case is the fact that the internal redundancy condition also references additional, non-local information. While we previously invoked external redundancy conditions in situations where the redundancy of an external element blocks spreading, the situation here is crucially different: the internal information (whether the prefix vowel under evaluation is $[-\text{high}]$) is combined with the non-local external information (whether the initial prefix vowel is $[+\text{high}]$) to redundantly specify the $[-\text{high}]$ prefix vowel under evaluation. The output [ATR] values of other prefix vowels are dependent on the output value of their successors, due to the spreading condition.

In order to facilitate the analysis, we define an input predicate $\text{initial/high}(x)$ in (35) which determines whether the initial vowel on a tier string is $[+\text{high}]$. This predicate recurses rightward through the input string and ‘spreads’ the information concerning whether the initial vowel of the string is $[+\text{high}]$ to all vowels to its right.

(35) Tutrugbu rightward-recursive initial $[+\text{high}]$ vowel input predicate

$\text{initial/high}(x) := \text{IF } \text{initial}(x) \text{ THEN } \text{high}(x) \text{ ELSE } \text{initial/high}(p(x))$

The [ATR] harmony function itself is provided in (36). The first line establishes root vowels as faithful spreading sources. The second line determines whether the domain element x is $[-\text{high}]$ — using the same $[-\text{high}](x)$ predicate as we used for Turkish — and if so, checks whether the form has an initial $[+\text{high}]$ vowel, using the rightward-recursive predicate just defined in (35). If this condition is also met, then the $[-\text{high}]$ vowel is redundantly $[-\text{ATR}]$ ($= \perp$); otherwise, the $[-\text{high}]$ vowel agrees with its successor ($= \phi_{\text{ATR}}(s(x))$). Finally, if the antecedent condition isn’t met because x is $[+\text{high}]$, then x also agrees with its successor.

(36) Tutrugbu [ATR] harmony function — inventory reduction blocking

$\phi_{\text{ATR}}(x) := \text{IF } \text{root}(x) \text{ THEN } \text{ATR}(x) \text{ ELSE}$
 $\quad \text{IF } [-\text{high}](x) \text{ THEN IF } \text{initial/high}(x) \text{ THEN } \perp \text{ ELSE } \phi_{\text{ATR}}(s(x))$
 $\quad \text{ELSE } \phi_{\text{ATR}}(s(x))$

The key to this function describing an unbounded circumambient pattern is that the recursions are CONTRADIRECTIONAL — the input predicate recurses rightward, the spreading function recurses leftward — and that they INTERACT:

²¹ Note that the $[-\text{high}, +\text{low}, -\text{ATR}]$ vowel [a] alternates with the $[-\text{high}, -\text{low}, +\text{ATR}]$ vowel [e] — yet another instance of ‘re-pairing’.

the leftward-recursive spreading function crucially relies on the information provided by the rightward-recursive input predicate (Heinz & Lai 2013, Jardine 2016, McCollum et al. 2020, to appear, Meinhardt et al. 2021, 2024).

We provide a BMRS evaluation table in Figure 6 to clarify our analysis of Tutrugbu. This table shows how the output [ATR] values of the vowels in [bɔbadiwu] are computed from the underlying [ATR] values of the vowels in /bɔ+ba+dɪ+√wu/ ‘1P+FUT+ITV+√climb’ (34f).²² The first block of predicates indicates all of the input values that are necessary to compute the conditions in the actual function, which appear in the next block. Domain element 4 satisfies the source faithfulness condition of being a root vowel (and [+ATR], conflated here). Its output value $\phi_{\text{ATR}}(x)$ is thus dependent on its input value $\text{ATR}(x)$, as indicated by the blue arrow. Domain element 1 satisfies the condition of being both initial and [+high], so the $\text{initial/high}(x)$ property is propagated rightward to all elements, as indicated by the red arrows. Domain element 2 thus satisfies the internal redundancy condition, meaning that its output [ATR] value is by default [-ATR], as indicated by the violet arrows. Finally, the outputs of domain elements 1 and 3 are dependent on the following vowels’ output values for [ATR]: domain element 3 precedes a [+ATR] vowel, so this value spreads from domain element 4; domain element 1 precedes a [-ATR] vowel, which is redundantly [-ATR] due to the internal redundancy condition, so this value spreads from domain element 2 to domain element 1.

Input	b	ɔ + b	a + d	i + √ w	u			
x	1	2	3		4			
root/ATR(x)	⊥	⊥	⊥		⊥			
high(x)	⊥	⊥	⊥		⊥			
initial(x)	⊥	⊥	⊥		⊥			
$[-\text{high}](x)$	⊥	⊥	⊥		⊥			
initial/high(x)	⊥	⊥	⊥		⊥			
$\phi_{\text{ATR}}(x)$	⊥	⊥	⊥	⊥	⊥			
Output	b	ɔ	b	a	d	i	w	u

Figure 6 Evaluation table for /bɔ+ba+dɪ+√wu/ → [bɔbadiwu] ‘1P+FUT+ITV+√climb’. Dashed arrows indicate the flow of computation and source of output values for each domain element.

It is worth comparing this BMRS analysis to the finite-state analysis provided by McCollum et al. (2020: 238ff), in which there are two subsequential finite-state transducers (FSTs): one rewriting the string leftward from the root and the other rewriting it rightward from the initial prefix. If the root vowel is [+ATR], then the leftward FST initiates spreading of [+ATR]. However, upon encountering the first [-high] vowel to the left of the root, the leftward FST specially encodes this vowel — and, recursively, all vowels further to the left — with information that the rightward FST can subsequently use to determine that there is a [+ATR] root vowel at the other end of the string. If the initial prefix vowel is [+high], the rightward FST outputs these specially encoded vowels as [-ATR] — this is the inventory reduction blocking effect — and if it is [-high], they are output as [+ATR]. This crucial interaction, with the leftward FST effectively deferring to the rightward FST upon encountering a [-high] vowel, mirrors the embedding of the rightward-recursive $\text{initial/high}(x)$ predicate in (35) within the IF $[-\text{high}](x)$ THEN ... redundancy condition of the overall leftward-recursive harmony function in (36). The special encoding of vowels by the leftward FST is akin to the information that is ‘spread’ across the input string by the rightward-recursive $\text{initial/high}(x)$ predicate; both kinds of information are like ‘breadcrumbs’ left behind by one component of the overall analysis to be followed by the other. The only difference is directional: the rightward FST is following the leftward FST’s breadcrumbs, while the leftward-recursive harmony function is following the rightward-recursive input predicate’s breadcrumbs. But this difference is entirely superficial, and in fact an interacting FST analysis that is more directly parallel to our BMRS analysis is possible: first, the rightward FST specially encodes all vowels if the first vowel in the string is [+high], and then the leftward FST spreads [+ATR] but accordingly rewrites those specially encoded vowels as [-ATR], starting from when a [-high] vowel is encountered (in a prefix, since this is inventory reduction blocking).²³

²² The surface [ATR] values of all prefix vowels are predictable, so their underlying values are arbitrary; we nevertheless specify them underlyingly as [-ATR]. McCollum et al. (2020: 222–224) adduce several pieces of evidence for the default value being [-ATR] in Tutrugbu.

²³ A similar direction reversal appears to be more complicated with BMRS, because we need to define two leftward-recursive input predicates: one that ‘spreads’ the information that the root vowel is [+ATR] — call this the ‘[+ATR] breadcrumb’ — and another that, starting from the rightmost [-high] prefix vowel, ‘spreads’ the information that there is at least one such vowel in the form — call this the ‘[-high] breadcrumb’. The output

Unbounded circumambient feature spreading patterns have received focused attention in the recent literature. They were effectively claimed to be unattested by Wilson (2003, 2006), but attestations later surfaced: at first with (arguably bounded) metaphony patterns (Walker 2010; cf. Mascaró 2019), then with tonal patterns (Hyman 2011, Jardine 2016), and later with unbounded harmony patterns (like Tutrugbu; McCollum & Essegbey 2018, McCollum et al. 2018, 2020, to appear). What is not in serious dispute is that unbounded circumambient patterns require greater computational expressivity than unbounded semiambient ones (Jardine 2016, Meinhardt et al. 2021, 2024, Yolyan 2025; cf. Lamont et al. 2019, O’Hara & Smith 2019, Smith & O’Hara 2019), and that unbounded circumambient patterns appear to be less commonly attested than unbounded semiambient ones (Jardine 2016, McCollum et al. 2018, 2020, to appear).

We’ve noted that the Tutrugbu function in (36) is an instance of unbounded circumambient inventory reduction blocking, with the source faithfulness conditions scoping over the recursive internal redundancy condition. Unbounded circumambient opaque blocking is also predicted to be possible, with the internal redundancy condition scoping over the source faithfulness condition, as is inventory expansion spreading, with the internal redundancy condition tucked in the THEN . . . clause of the source faithfulness condition. The internal redundancy condition can also be changed to an external one, leading to icy target blocking, or augmented with an external one, resulting in full blocking. The fact that unbounded circumambient patterns are relatively uncommon makes it a particular challenge to identify examples of these predicted patterns, a challenge we hope will be taken up in future research building on this work.

5.3 Summary

In this section we have followed through on two theoretical consequences of the BMRS approach to the typology of feature spreading and blocking discussed in the previous two sections. The first consequence concerns the relatively minor role that particularities of representation plays in our approach, compared to the relatively more prominent role that representation has played in accounts of feature spreading and blocking since the mid-1970s. We contend that these representational approaches nevertheless require often substantial computational structure both to determine and to interpret those representations, a fact that is particularly obvious when the language-particular differences in blocking that form points in our typology are taken into account. This begs the question whether the representational assumptions are independently needed, beyond their graphically satisfying qualities. Leaning into the independently necessary computational tools, as we have here, provides a direct method for deriving the typology.

The second consequence concerns the relation between our typology and computationally more expressive, bidirectional patterns, and we have shown how the general structure of the typology runs orthogonal to the expressivity dimension. Bidirectional patterns require recursion in both directions, but this does not change the overall characterization of those patterns within our typology: the unbounded semiambient pattern in Degema is a case of bidirectional full spreading, the unbounded circumambient pattern in Tutrugbu is a case of bidirectional inventory reduction blocking, and there exist patterns instantiating other points in this part of the typological space (e.g., full spreading in one direction but some form of blocking in the other, as in Maasai and Palestinian Arabic). More work needs to be done to describe new patterns in this space and/or to match existing descriptions with points in this space, and our characterization of the typology of spreading and blocking patterns provides a structured roadmap for exploring it.

6 Final remarks

In this paper we have defined the basic substantive elements of feature spreading and blocking with BMRS — source faithfulness conditions, spreading conditions, and internal and external redundancy conditions — and showed how they can be deployed to generate a structured typology of attested feature spreading and blocking patterns. This demonstration illustrates one way in which a substantive theory of possible structures in BMRS can be developed. First, the relationship between the output property being computed by a harmony function and the redundancy conditions that form a subpart of that function may be restricted by substantive phonetic properties (see e.g. Archangeli & Pulleyblank 1994). Second, we have shown how the presence and scope of redundancy conditions can substantively restrict the way different BMRS structures interact. Both of these play a role in understanding our typology more broadly.

[ATR] specifications of all vowels are then determined by a rightward-recursive function that follows both types of breadcrumbs. If the initial vowel has no [+ATR] breadcrumb, then all vowels are output as [−ATR]. If the initial vowel *does* have an [+ATR] breadcrumb, then it depends: (i) if it does *not* also have a [−high] breadcrumb, then all vowels are output as [+ATR]; (ii) if it *does* also have a [−high] breadcrumb, and the initial vowel is itself [−high], then all vowels are again output as [+ATR]; (iii) otherwise (i.e., if the initial vowel is [+high] and has both breadcrumbs), then all vowels with the [−high] breadcrumb are output as [−ATR], and remaining vowels to the right are output as [+ATR].

Existing alternative approaches struggle to describe some pattern or other in this typology, consequently failing to characterize it in the structured way that we have here. Representational approaches, which for the past half century have dominated the study of this area of phonology, require a number of additional mechanisms in order to move beyond the classic full spreading / opaque blocking dyad: inventory expansion spreading is anti-structure preserving (Kiparsky 1981); inventory reduction blocking is contrast licensing in privileged (source) positions (Kaplan 2015, Kaplan & Walker 2024); full blocking is due to harmony being limited to a subset of vowels (e.g. to non-high vowels in the case of Halh rounding harmony; Steriade 1979, van der Hulst & Smith 1987, Svantesson et al. 2005, van der Hulst 2018). Icy target blocking has received a number of different analyses involving metrical structure (Jurgec 2011, McCollum 2022), additional local assimilation rules (van der Hulst 2018), and contrast preservation (Ozburn & Leduc 2022).

The computational approach we have taken is largely agnostic to specific representational choices. It is compatible with any theory of representation maintaining a skeletal tier of some kind, thus moving us closer to identifying the components that are required for any system to compute the types of input-output maps that have been discussed throughout this paper. While these components may take on slightly different forms depending on the specific theory they are embedded within, their function at an abstract level remains the same. Our approach provides a vocabulary for explaining *why* further assumptions are needed to describe certain types of patterns within other approaches. For example, the distinction between inventory expansion and reduction is the result of varying the scope of an internal redundancy condition; in order to compute these types of patterns, any theory needs some way to differentiate between how output forms are computed in different positions. There are various ways to accomplish this, at least one of which is needed in any theory if we want to ensure these types of patterns have formal solutions. Our computational typology therefore has the capacity not only to describe, but also to diagnose at a metatheoretical level.

One issue with this type of grammatically-driven typological work is determining whether the unattestedness of a given predicted pattern is due to formal grammatical factors or some other, external factors (e.g., phonetics, sound change, learnability, etc.). As we have remarked on several times throughout this paper, we believe that the interaction between redundancy and faithfulness conditions provides a new roadmap for exploring spreading and blocking patterns in interesting ways. For example, we now have a specific computational property — external redundancy — which seems to be a component of both the typologically rare and unattested patterns. One advantage of novel theoretical approaches is that they are able to shine a light on previously unrecognized properties of linguistic systems. We are optimistic that if the approach we have pursued is embraced, then even more new insights are likely to emerge.

References

- Aksënova, Alëna, Jonathan Rawski, Thomas Graf & Jeffrey Heinz. 2024. The computational power of harmonic forms. In Nancy A. Ritter & Harry van der Hulst (eds.), *The Oxford Handbook of Vowel Harmony*, chap. 34, 437–451. Oxford University Press. doi:10.1093/oxfordhb/9780198826804.013.34.
- Archangeli, Diana & Douglas Pulleyblank. 1994. *Grounded Phonology*. MIT Press.
- Archangeli, Diana & Douglas Pulleyblank. 2007. Harmony. In Paul de Lacy (ed.), *The Cambridge Handbook of Phonology*, 353–378. Cambridge University Press.
- Awobuluyi, A. Oladele. 1967. Vowel and consonant harmony in Yoruba. *Journal of African Languages* 6. 1–8.
- Baković, Eric. 2000. *Harmony, dominance and control*: Rutgers University dissertation.
- Bamgboṣe, Ayọ. 1967. Vowel harmony in Yoruba. *Journal of African Languages* 6. 268–273.
- Bhaskar, Siddharth, Jane Chandlee & Adam Jardine. 2023. Rational functions via recursive schemes. ArXiv preprint arXiv:2302.03074.
- Bhaskar, Siddharth, Jane Chandlee, Adam Jardine & Christopher Oakden. 2020. Boolean monadic recursive schemes as a logical characterization of the subsequential functions. In Alberto Leporati, Carlos Martín-Vide, Dana Shapira & Claudio Zandron (eds.), *Language and Automata Theory and Applications Lecture Notes in Computer Science*, 157–169. Springer. doi:10.1007/978-3-030-40608-0_10.

- Burness, Phillip, Kevin McMullin & Andrew Nevins. 2024. Revisiting locality in vowel harmony. In Nancy A. Ritter & Harry van der Hulst (eds.), *The Oxford Handbook of Vowel Harmony*, chap. 22, 269–290. Oxford University Press. doi:10.1093/oxfordhb/9780198826804.013.22.
- Burness, Phillip Alexander, Kevin James McMullin & Jane Chandlee. 2021. Long-distance phonological processes as tier-based strictly local functions. *Glossa: a journal of general linguistics* 6(1). doi:10.16995/glossa.5780.
- Casali, Roderic F. 2012. [+ATR] dominance in Akan. *Journal of West African Languages* 39(1). 33–59.
- Chandlee, Jane & Adam Jardine. 2021. Computational universals in linguistic theory: Using recursive programs for phonological analysis. *Language* 97(3). 485–519. doi:10.1353/lan.2021.0045.
- Chomsky, Noam & Morris Halle. 1968. *The Sound Pattern of English*. Harper & Row.
- Clements, George N. 1981. Akan vowel harmony: a nonlinear analysis. In *Harvard Studies in Phonology* 2, 108–177.
- Clements, George N. & Engin Sezer. 1982. Vowel and consonant disharmony in Turkish. In Harry van der Hulst & Norval Smith (eds.), *The Structure of Phonological Representations, Pt. 2*, 213–255. Foris.
- Davis, Stuart. 1995. Emphasis spread in Arabic and Grounded Phonology. *Linguistic Inquiry* 26. 465–498.
- De Santo, Aniello & Thomas Graf. 2019. Structure sensitive tier projection: Applications and formal properties. In *Formal Grammar: 24th International Conference, FG 2019, Riga, Latvia, August 11, 2019, Proceedings 24*, 35–50. Springer.
- Dolphyne, Florence Abena. 1988. *The Akan (Twi-Fante) language: Its sound systems and tonal structure*. Accra: Ghana Universities Press.
- Goldsmith, John. 1985. Vowel Harmony in Khalkha Mongolian, Yaka, Finnish and Hungarian. *Phonology Yearbook* 2. 253–275. <https://www.jstor.org/stable/4419959>.
- Goldsmith, John A. 1976. *Autosegmental Phonology*: MIT dissertation.
- Graf, Thomas. 2016. Weak determinism is not too weak: A reply to Jardine (2016) Unpublished ms., Stony Brook University.
- Greenberg, Joseph H. 1966. Synchronic and diachronic universals in phonology. *Language* 42(2). 508–517.
- Halle, Morris. 1959. *The Sound Pattern of Russian*. Mouton.
- Hao, Y. & S. Andersson. 2019. Unbounded Stress in Subregular Phonology. In *SIGMORPHON 16*, 135–143. ACL.
- Heinz, Jeffrey. 2018. The computational nature of phonological generalizations. In Larry M. Hyman & Frans Plank (eds.), *Phonological Typology*, 126–195. De Gruyter Mouton. doi:10.1515/9783110451931-005.
- Heinz, Jeffrey & Regine Lai. 2013. Vowel harmony and subsequentiality. In *Proceedings of the 13th Meeting on the Mathematics of Language (MoL 13)*, 52–63.
- Heinz, Jeffrey, Chetan Rawal & Herbert G. Tanner. 2011. Tier-based strictly local constraints for phonology. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics*, 58–64.
- van der Hulst, Harry. 2018. *Asymmetries in Vowel Harmony: A Representational Account*. Oxford University Press.
- van der Hulst, Harry & Norval Smith. 1987. Vowel harmony in Khalkha and Buriat (East Mongolian). In Frits Beukema & Peter Coopmans (eds.), *Linguistics in the Netherlands 1987*, 81–91. Dordrecht: Foris.
- Hyman, Larry. 2008. Universals in phonology. *The Linguistic Review* 25. 83–137. doi:10.1515/TLIR.2008.003.
- Hyman, Larry M. 2011. Tone: is it different? In John A. Goldsmith, Jason Riggle & Alan C. L. Yu (eds.), *The Handbook of Phonological Theory, 2nd edition*, 197–239. Wiley-Blackwell.

- Hyman, Larry M. & Frans Plank (eds.). 2018. *Phonological Typology*, vol. 23 Phonetics and Phonology. de Gruyter Mouton.
- Jardine, Adam. 2016. Computationally, tone is different. *Phonology* 33(2). 247–283.
- Jardine, Adam & Christopher Oakden. 2023. Computing process-specific constraints. *Linguistic Inquiry* 1–9. doi: 10.1162/ling_a_00510.
- Johnson, C. Douglas. 1972. *Formal Aspects of Phonological Description*. De Gruyter Mouton.
- Joos, Martin. 1942. A phonological dilemma in Canadian English. *Language* 141–144.
- Jurgec, Peter. 2011. *Feature spreading 2.0: A unified theory of assimilation*: Universitetet i Tromsø dissertation.
- Kaplan, Aaron. 2015. Maximal prominence and a theory of possible licensors. *Natural Language & Linguistic Theory* 33. 1235–1270. doi:10.1007/s11049-014-9273-5.
- Kaplan, Aaron & Rachel Walker. 2024. What constitutes privileged positions in vowel harmony? In Nancy A. Ritter & Harry van der Hulst (eds.), *The Oxford Handbook of Vowel Harmony*, chap. 23, 291–314. Oxford University Press. doi:10.1093/oxfordhb/9780198826804.013.23.
- Kaplan, Ronald M & Martin Kay. 1994. Regular models of phonological rule systems. *Computational linguistics* 20(3). 331–378.
- Keri, Ethelbert E. 1997. *Degema*. München-Newcastle: Lincom Europa.
- Kiparsky, Paul. 1981. Vowel harmony. Unpublished manuscript, MIT.
- Koser, Nate & Adam Jardine. 2020. The computational nature of stress assignment. In *Proceedings of the 2019 Annual Meeting on Phonology*, doi:10.3765/amp.v8i0.4676.
- Kpogo, Felix. 2024. *Investigating sound change in Twi vowel harmony: a sociophonetic study of age, gender, and locality effects*: Boston University dissertation.
- Lamont, Andrew, Charlie O’Hara & Caitlin Smith. 2019. Weakly deterministic transformations are subregular. In *Proceedings of the 16th Workshop on Computational Research in Phonetics, Phonology, and Morphology*, 196–205.
- Mascaró, Joan. 2019. On the lack of evidence for nonmyopic harmony. *Linguistic Inquiry* 50(4). 862–872. doi: 10.1162/ling_a_00327.
- Mayer, Connor & Travis Major. 2018. A challenge for tier-based strict locality from Uyghur backness harmony. In Annie Foret, Greg Kobele & Sylvain Pogodalla (eds.), *Formal Grammar 2018 Lecture Notes in Computer Science*, vol 10950, Berlin, Heidelberg: Springer. doi:10.1007/978-3-662-57784-4_4.
- McCarthy, John J. 1997. Process-specific constraints in Optimality Theory. *Linguistic Inquiry* 28. 231–251.
- McCarthy, John J. 2008. *Doing Optimality Theory: Applying theory to data*. John Wiley & Sons.
- McCarthy, John J. & Alison Taub. 1992. Review of C. Paradis and J.-F. Prunet, eds. (1991), *The Special Status of Coronals*. *Phonology* 9. 363–370.
- McCollum, Adam G. 2022. Non-iterativity, icy targets, and the need for non-linear representations in feature spreading. In *Proceedings of the 2021 Annual Meeting on Phonology*, doi:10.3765/amp.v9i0.5205.
- McCollum, Adam G., Eric Baković & Anna Mai. to appear. On the inevitability of non-myopic harmony. *Phonology* (Special thematic issue: “Metaphony and Umlaut: Theoretical Issues,” ed. by Michela Russo and Rachel Walker).
- McCollum, Adam G., Eric Baković, Anna Mai & Eric Meinhardt. 2018. The expressivity of segmental phonology and the definition of weak determinism. Ms., UC San Diego, available at <https://ling.auf.net/lingbuzz/004197>.

- McCollum, Adam G., Eric Baković, Anna Mai & Eric Meinhardt. 2020. Unbounded circumambient patterns in segmental phonology. *Phonology* 37(2). 215–255.
- McCollum, Adam G. & James Essegbey. 2018. Unbounded harmony is not always myopic: evidence from Tutrugbu. In *Proceedings of the 35th West Coast Conference on Formal Linguistics*, 251–258. Cascadia Proceedings Project.
- McCollum, Adam G. & Adam Jardine. 2022. Input- and output-oriented generalizations in Iny ATR harmony. Ms., Rutgers University, available at <https://ling.auf.net/lingbuzz/006830>.
- Meinhardt, Eric, Anna Mai, Eric Baković & Adam McCollum. 2024. Weak determinism and the computational consequences of interaction. *Natural Language & Linguistic Theory* 42(3). 1191–1232.
- Meinhardt, Eric, Anna Mai, Eric Baković & Adam G. McCollum. 2021. On the proper treatment of weak determinism: Subsequentiality and simultaneous application in phonological maps. In *San Diego Linguistics Papers* 9, <https://escholarship.org/uc/item/6f20b9xx>.
- Nelson, Scott & Eric Baković. 2024a. Underspecification without underspecified representations. *Proceedings of the Society for Computation in Linguistics* 7(1). 352–356. doi:10.7275/scil.2227.
- Nelson, Scott & Eric Baković. 2024b. Computation clarifies mixed specification behavior of coronals in English. Poster presented at the 2024 Annual Meeting on Phonology, Rutgers University.
- Oakden, Christopher. 2021. *Modeling phonological interactions using recursive schemes*: Rutgers University dissertation.
- O’Hara, Charlie & Caitlin Smith. 2019. Computational complexity and sour-grapes-like patterns. In *Proceedings of the 2018 Annual Meeting on Phonology*, doi:10.3765/amp.v7i0.4502.
- Ozburn, Avery & Marjorie Leduc. 2022. Icy Targets in Karajá ATR Harmony as Contrast Preservation. In *Proceedings of the 2021 Annual Meeting on Phonology*, doi:10.3765/amp.v9i0.5190.
- Padgett, Jaye. 2002. Russian voicing assimilation, final devoicing, and the problem of [v]. ROA-528, Rutgers Optimality Archive, <https://roa.rutgers.edu/>.
- Pater, Joe. 1999. Austronesian nasal substitution and other NC effects. In René Kager, Harry van der Hulst & Wim Zonneveld (eds.), *The Prosody-Morphology Interface*, 310–343. Cambridge: Cambridge University Press. ROA-160, Rutgers Optimality Archive, <http://roa.rutgers.edu>.
- Prince, Alan & Paul Smolensky. 1993/2004. *Optimality Theory: Constraint Interaction in Generative Grammar*. John Wiley & Sons.
- Quinn-Wriedt, Lindsey Taylor. 2013. *Vowel Harmony in Maasai*: University of Iowa dissertation.
- Ribeiro, Eduardo Rivail. 2002. Directionality in vowel harmony: The case of Karajá (Macro-Jê). In *Proceedings of the Annual Meeting of the Berkeley Linguistics Society*, 475–485.
- Ribeiro, Eduardo Rivail. 2012. *A grammar of Karajá*: University of Chicago dissertation.
- Ringen, Catherine O. 1975. *Vowel harmony: theoretical implications*: Indiana University dissertation. Published by Garland, New York, 1988.
- Sapir, David. 1965. *A Grammar of Diola-Fogny*. Cambridge: Cambridge University Press.
- Smith, Caitlin & Charlie O’Hara. 2019. Formal characterizations of true and false sour grapes. In *Proceedings of the Society for Computation in Linguistics*, vol. 2 1, 338–341.
- Steriade, Donca. 1979. Vowel harmony in Khalkha Mongolian. In Ken Safir (ed.), *Papers on Syllable Structure, Metrical Structure, and Harmony Processes* (MIT Working Papers in Linguistics 1), 25–50.
- Svantesson, Jan-Olof. 1985. Vowel harmony shift in Mongolian. *Lingua* 67. 283–327.

- Svantesson, Jan-Olof. 2024. Vowel harmony in mongolic languages. In Nancy A. Ritter & Harry van der Hulst (eds.), *The Oxford Handbook of Vowel Harmony*, chap. 60, 774–779. Oxford University Press. doi: 10.1093/oxfordhb/9780198826804.013.60.
- Svantesson, Jan-Olof, Anna Tsendina, Anastasia Karlsson & Vivan Franzén. 2005. *The phonology of Mongolian*. Oxford University Press.
- Walker, Rachel. 2010. Nonmyopic harmony and the nature of derivations. *Linguistic Inquiry* 41(1). 169–179.
- Wilson, Colin. 2003. Analyzing unbounded spreading with constraints: marks, targets, and derivations. Unpublished manuscript, UCLA.
- Wilson, Colin. 2006. Unbounded spreading is myopic. Talk presented at the Phonology Fest Workshop on Current Perspectives on Phonology, Indiana University.
- Yolyan, Tatevik. 2025. A logical characterization of weak determinism as simultaneous application. *Journal of Logic, Language and Information* 1–65.